# SOCIOL 723: Social Statistics II

## Week 10, Panel Data and Difference-in-Differences

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# Today

Fixed Effects and Random Effects, Without Time

Panel Data and Unobserved Heterogeneity

Difference-in-Differences

The Trouble with Two-Way Fixed Effects

Beyond a Binary, Absorbing Treatment

Applications in Sociology

Practice

★ = essential, you must understand this ★ = for understanding, not examined

# A Third Source of Variation

So far: adjust for observables (Week 6), find an instrument (Week 7), find a threshold (Week 9). Today: use variation over time within the same unit.

## The core idea

If we observe the same unit before and after, we can difference away everything about that unit that does not change, including confounders we never measured and could not name.

- That is a genuinely powerful trick, and it is why panel data are prized.
- It buys nothing against confounders that *do* change over time, and the price is a new and demanding assumption about counterfactual trends.
- The last decade has shown that the standard implementation, two-way fixed effects, is badly behaved in exactly the setting sociologists use most.

# Plan

**Goal**: what fixed effects and difference-in-differences estimate, under which assumptions, and how to estimate it when the textbook regression fails.

1. **Fixed and random effects without time**: students in schools; Hausman, Mundlak.
2. **Panel fixed effects**: two-way FE, strict exogeneity, FE vs FD, clustering, dynamic panels; the marriage wage premium on real data.
3. **The** $2 \times 2$: parallel trends, scale, covariates (three estimators), inference with serial correlation.
4. **Event studies**: what pre-trend tests can and cannot show; base periods; sensitivity to parallel-trends violations.
5. **Staggered adoption**: why TWFE fails; the decomposition; the new estimators; all of them on one dataset (divorce laws and suicide).
6. **Beyond the binary absorbing treatment**: doses, switching on and off, triple differences, distributions, design-based views.
7. **Applications in sociology**.

Running data: NLSY men 1980–87 (`wooldridge::wagepan`); Stevenson–Wolfers state panel 1964–96 (`bacondecomp::divorce`).

# Fixed Effects

# Fixed Effects: Goal and Plan

**Goal**: compare each unit only with other units *in the same group*, so that nothing the group members share can confound the comparison.

| group | member | what the group effect absorbs |
|---|---|---|
| school | student | resources, teachers, neighborhood |
| family | sibling | parents, home, family income |
| firm | worker | pay policy, industry, location |
| person | year of that person's life | ability, upbringing (next section) |

1. The model, and why pooled OLS is biased.
2. Route 1: a separate intercept for each group. Route 2: subtract group means.
3. Why the two routes agree, and which variation they use.
4. Random effects, and testing fixed against random.

Data: High School and Beyond, 7,185 students in 160 schools (Week 4) (figs/w10_fe.R).

# Step 1: The Model and the Problem

Student $i$ in school $j$; $Y_{ij}$ math score, $X_{ij}$ family SES:

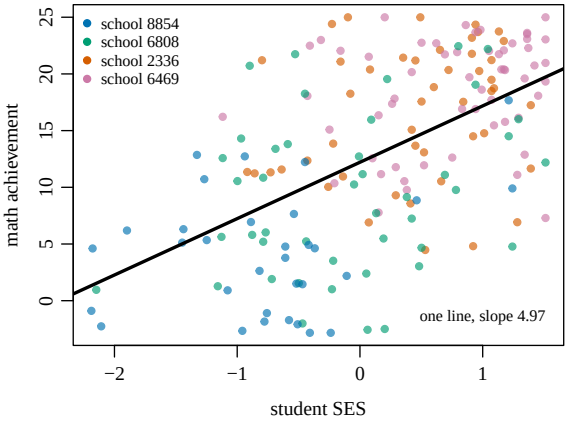
$$Y_{ij} = \beta X_{ij} + \alpha_j + \epsilon_{ij}, \qquad \alpha_j = \text{everything about school } j \text{ that all its students share.}$$

Pooled OLS of $Y_{ij}$ on $X_{ij}$ leaves $\alpha_j$ in the error. By the omitted-variable formula (Week 1):

$$\operatorname{plim} \hat{\beta}_{\text{pooled}} = \beta + \frac{\operatorname{Cov}(X_{ij}, \alpha_j)}{\operatorname{Var}(X_{ij})} \qquad \alpha_j \text{ is the omitted variable.}$$

- High-SES students attend schools with more resources: $\operatorname{Cov}(X_{ij}, \alpha_j) > 0$, so the pooled slope is too large.
- HSB: pooled 3.18 (0.13); comparing students only with schoolmates, 2.19 (0.13). Clustered by school.

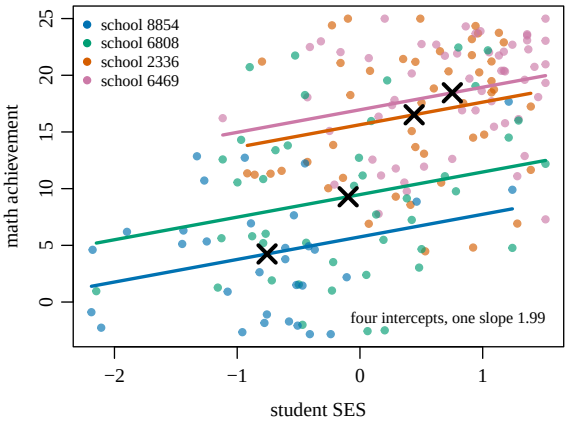
# Pooled OLS on Four Schools




Four HSB schools, 180 students. School mean SES $-0.76$, $-0.10$, $0.44$, $0.75$; mean math 4.2, 9.3, 16.5, 18.5.

- One line through all students mixes two comparisons: students against their schoolmates, and schools against schools.
- The second comparison is driven by $\alpha_j$: the high-SES schools are also the high-math schools.
- Slope 4.97 on these four schools.

# Step 2, Route 1: A Separate Intercept for Each School




Treat each $\alpha_j$ as a parameter: one dummy per school.

$$Y_{ij} = \beta X_{ij} + \sum_k \alpha_k \mathbb{1}\{j = k\} + \epsilon_{ij}$$

- Each school gets its own height; all schools share one slope, 1.99.
- Each line passes through its school's means ($\times$): $\hat{\alpha}_j = \bar{Y}_j - \hat{\beta}\bar{X}_j$, the usual intercept formula, school by school.
- The difference in heights is no longer credited to SES.

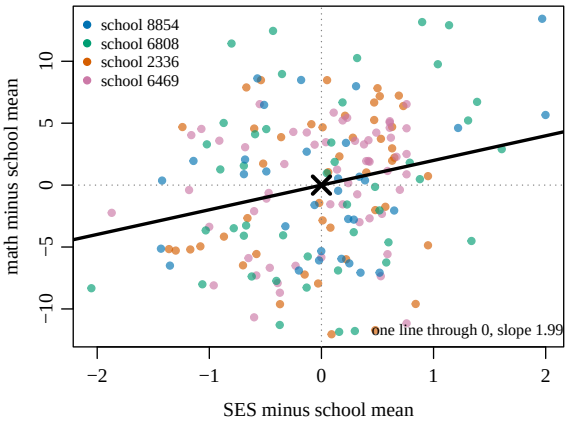
# Step 2, Route 2: Subtract the School Means

Average the model over the $n_j$ students of school $j$, then subtract:

$$
\begin{aligned}
Y_{ij} &= \beta X_{ij} + \alpha_j + \epsilon_{ij} && \text{the model} \\
\bar{Y}_j &= \beta \bar{X}_j + \alpha_j + \bar{\epsilon}_j && \text{average over } i \text{ in school } j;\ \alpha_j \text{ is the same for all } i \\
Y_{ij} - \bar{Y}_j &= \beta \left(X_{ij} - \bar{X}_j\right) + \left(\epsilon_{ij} - \bar{\epsilon}_j\right) && \text{subtract: } \alpha_j - \alpha_j = 0
\end{aligned}
$$

- $\alpha_j$ is gone without being named or measured.
- OLS of $Y_{ij} - \bar{Y}_j$ on $X_{ij} - \bar{X}_j$, no intercept, estimates $\beta$ from within-school differences only.
- The question becomes: is a student with more SES *than her schoolmates* also higher in math *than her schoolmates*?

# What Demeaning Does to the Picture




Same 180 students after subtracting their school's means.

- Every school's $\times$ moves to $(0, 0)$; the four clouds sit on top of each other.
- The between-school spread of the first picture is deleted; only spread *inside* each school is left.
- One line through the origin, slope 1.99: the same number as four intercepts.

# Step 3: Why the Two Routes Agree

**FWL (Week 1)**: the coefficient on $X$ in a regression with dummies equals the slope from regressing $Y$-net-of-dummies on $X$-net-of-dummies.

$$\begin{aligned} \text{fit of } X_{ij} \text{ on dummies} &= \underset{c_1,\ldots,c_J}{\arg\min} \textstyle\sum_j \sum_i (X_{ij} - c_j)^2 && \text{one constant per school} \\ &= \bar{X}_j && \text{the mean minimizes squared deviations} \\ X_{ij} \text{ net of dummies} &= X_{ij} - \bar{X}_j && \text{the residual; same for } Y_{ij} \end{aligned}$$

| HSB, all 160 schools | SES | (se) |
|---|---|---|
| 160 school dummies, `lm` | 2.1912 | (0.1086) |
| demeaned by hand, `lm` | 2.1912 | (0.1074) |
| `feols(math ~ SES \| School)`, clustered | 2.1912 | (0.1298) |

By hand, `lm` counts $N - 1$ degrees of freedom; 160 means were estimated, so the truth is $N - 1 - 160$. Small here (1%); with siblings ($n_j = 2$) the by-hand s.e. is too small by a factor of about $\sqrt{2}$. Use FE software.

# Step 3: Which Variation Fixed Effects Uses

Split each student's SES into the school's part and her own part:

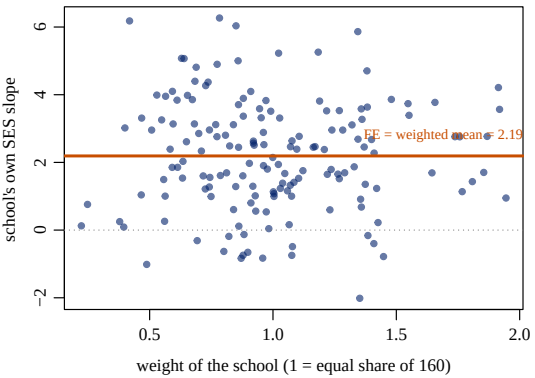
$$X_{ij} - \bar{X} = \underbrace{(\bar{X}_j - \bar{X})}_{\text{between}} + \underbrace{(X_{ij} - \bar{X}_j)}_{\text{within}} \qquad \text{add and subtract } \bar{X}_j$$

$$\textstyle\sum_{ij}(X_{ij} - \bar{X})^2 = \sum_{ij}(\bar{X}_j - \bar{X})^2 + \sum_{ij}(X_{ij} - \bar{X}_j)^2 \qquad \text{cross term is 0}$$

$$4{,}364 = 1{,}229 + 3{,}135 \qquad \text{HSB SES}$$

- FE uses only the 72% of SES variation that lies within schools; the 28% between schools is discarded.
- A group with no within variation contributes nothing: siblings with the same value, or a school where every student has the same SES.
- Precision depends on within variation. Here FE removes $\alpha_j$ from the error too, so its s.e. (0.130) is close to pooled OLS's (0.134).

# Fixed Effects Averages the Schools' Own Slopes



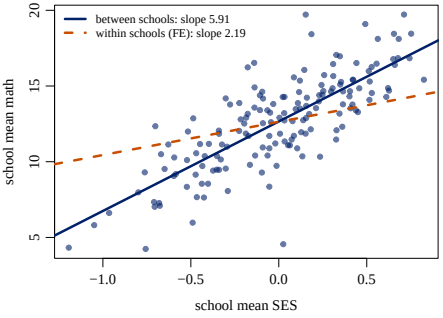

With $\tilde{X}_{ij} = X_{ij} - \bar{X}_j$ and $\tilde{Y}_{ij} = Y_{ij} - \bar{Y}_j$:

$$\hat{\beta}_{FE} = \frac{\sum_j \sum_i \tilde{X}_{ij}\tilde{Y}_{ij}}{\sum_j \sum_i \tilde{X}_{ij}^2} = \sum_j w_j \, b_j$$

$b_j = \sum_i \tilde{X}_{ij}\tilde{Y}_{ij} / \sum_i \tilde{X}_{ij}^2$ is school $j$'s own slope; $w_j = \sum_i \tilde{X}_{ij}^2 / \sum_k \sum_i \tilde{X}_{ik}^2$ (multiply and divide by $\sum_i \tilde{X}_{ij}^2$).

- 160 slopes from $-2.02$ to 6.27; weighted mean 2.19.
- Weight grows with school size and with SES spread inside the school: from 0.22 (20 students) to 1.94 (53 students) times an equal share.

# The Comparison Fixed Effects Throws Away




The **between** estimator regresses the 160 school means $\bar{Y}_j$ on $\bar{X}_j$: slope 5.91 (0.37), almost three times the within slope.

Pooled OLS mixes the two. Let $W$, $B$ be the within and between sums of squares of SES (previous frame), and $b_W$, $b_B$ the within and student-weighted between slopes.

$$\hat{\beta}_{\text{pooled}} = \frac{Wb_W + Bb_B}{W + B}$$

split the covariance, like the variance

$$= \frac{3{,}135(2.19) + 1{,}229(5.72)}{4{,}364} = 3.18$$

HSB

# What Fixed Effects Cannot Do

- **No school-level variables.** Catholic is the same for every student in a school: $\text{Catholic}_j - \overline{\text{Catholic}}_j = 0$. With 160 dummies, `lm` returns `NA` for Catholic.
- **A narrower question.** $\hat{\beta}_{FE}$ is the SES gap among schoolmates. It says nothing about moving a student to a higher-SES school.
- **Only what is shared is removed.** A confounder that varies among schoolmates (a parent's involvement) still biases $\hat{\beta}_{FE}$.
- **Noise looms larger.** Measurement error in SES is not shared by schoolmates, so it survives demeaning while part of the signal does not: attenuation is worse within than pooled.

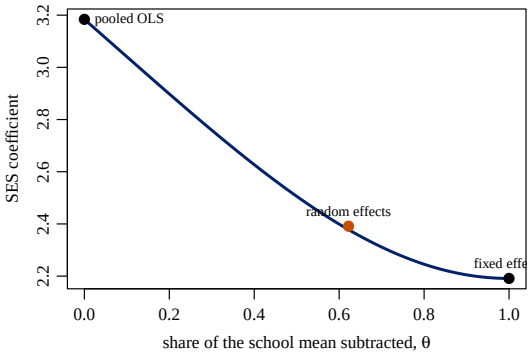
# Step 4: Random Effects

Same equation, but $\alpha_j = \mu + u_j$ is treated as a random draw (Week 4's random intercept):

$$Y_{ij} = \mu + \beta X_{ij} + u_j + \epsilon_{ij}, \qquad u_j \sim \mathcal{N}(0, \tau^2), \qquad E[u_j \mid X_{1j}, \ldots, X_{n_j j}] = 0.$$

- The last condition is the price: the school effect is unrelated to the SES of *every* student in the school. Pooled OLS needs the same.
- If it holds, RE uses between and within variation, so it is more efficient, and it estimates school-level variables: Catholic 2.10 (0.34).
- If it fails, RE is biased, as pooled OLS is.
- Fixed effects asks nothing about $\alpha_j$: it may correlate with SES in any way.

# Random Effects Is Partial Demeaning




RE is OLS after subtracting only a *share* $\theta_j$ of each school mean:

$$Y_{ij} - \theta_j \bar{Y}_j = \beta(X_{ij} - \theta_j \bar{X}_j) + \ldots$$

$$\theta_j = 1 - \sqrt{\frac{\sigma^2}{\sigma^2 + n_j \tau^2}}$$

- $\tau^2 = 0$: $\theta_j = 0$, pooled OLS. $n_j \tau^2$ large: $\theta_j \to 1$, FE.
- HSB: $\hat{\tau}^2 = 4.73$, $\hat{\sigma}^2 = 37.0$; $\theta_j$ from 0.40 ($n_j = 14$) to 0.68 ($n_j = 67$).
- OLS on the quasi-demeaned data: 2.3915 = `lmer`'s 2.3915.

# Four Estimators of One Coefficient

| estimator | subtracts from each school | uses | SES (se) |
|---|---|---|---|
| pooled OLS | nothing | all variation | 3.18 (0.13) |
| between | (uses the 160 means only) | between schools | 5.91 (0.37) |
| random effects | $\theta_j$ of the mean, 0.40–0.68 | a weighted mix | 2.39 (0.11) |
| fixed effects | the whole mean | within schools | 2.19 (0.13) |

- The less of the school mean we subtract, the more the estimate leans on the between comparison, and the larger it gets.
- RE sits close to FE here because schools are large ($n_j \approx 45$), so $\theta_j$ is large. With siblings ($n_j = 2$) and the same variances, $\theta = 0.11$: RE would sit close to pooled OLS.
- Is the gap 2.39 vs 2.19 more than noise? That is the test.

Pooled, between, FE: clustered by school. RE: model-based, `lmer`, ML.

# Step 4: The Hausman Test, the Logic (Hausman 1978)

$H_0$: $E[u_j \mid X] = 0$.

| | $H_0$ true | $H_0$ false |
|---|---|---|
| fixed effects | consistent | consistent |
| random effects | consistent, efficient | inconsistent |

A large gap $\hat{\beta}_{FE} - \hat{\beta}_{RE}$ is evidence against $H_0$. To scale it:

$$\begin{aligned} \mathrm{Var}(\hat{\beta}_{FE}) &= \mathrm{Var}\left(\hat{\beta}_{RE} + (\hat{\beta}_{FE} - \hat{\beta}_{RE})\right) && \text{add and subtract } \hat{\beta}_{RE} \\ &= \mathrm{Var}(\hat{\beta}_{RE}) + \mathrm{Var}(\hat{\beta}_{FE} - \hat{\beta}_{RE}) && \text{under } H_0,\ \hat{\beta}_{RE} \text{ is uncorrelated with the gap} \end{aligned}$$

$$H = \frac{(\hat{\beta}_{FE} - \hat{\beta}_{RE})^2}{\mathrm{Var}(\hat{\beta}_{FE}) - \mathrm{Var}(\hat{\beta}_{RE})} \xrightarrow{d} \chi^2_1 \quad K \text{ coefficients: the matrix version, } \chi^2_K$$

Why uncorrelated: if the gap moved with $\hat{\beta}_{RE}$, adding a bit of the gap to $\hat{\beta}_{RE}$ would give a consistent estimator with smaller variance, so $\hat{\beta}_{RE}$ was not efficient.

# The Hausman Test on HSB

$$\hat{\beta}_{FE} - \hat{\beta}_{RE} = 2.1912 - 2.3915 = -0.2003 \qquad \text{the gap}$$

$$\mathrm{Var}(\hat{\beta}_{FE}) - \mathrm{Var}(\hat{\beta}_{RE}) = 0.10865^2 - 0.10569^2 = 0.000633 \qquad \text{model-based s.e.s}$$

$$H = 0.2003^2/0.000633 = 63.4, \quad p < 0.001 \qquad \chi^2_1$$

Reject: SES is correlated with the school effect, and RE's 2.39 is biased.

- The denominator is a small difference between two similar numbers; in other samples it can be negative, and $H$ is then undefined.
- It requires RE to be efficient: homoskedastic errors, no correlation within school beyond $u_j$. Clustered s.e.s break the formula.
- It depends on how $\tau^2$ is estimated: `plm`'s default RE gives $H = 112$.
- Not rejecting does not show $H_0$ holds; it may be low power.

# A Robust Version: Mundlak's Regression (Mundlak 1978)

Let the school effect depend on the school's mean SES: $\alpha_j = \mu + \gamma \bar{X}_j + a_j$, $a_j$ unrelated to $X$.

$$\begin{aligned} Y_{ij} &= \beta X_{ij} + \alpha_j + \epsilon_{ij} && \text{the FE model} \\ &= \mu + \beta X_{ij} + \gamma \bar{X}_j + (a_j + \epsilon_{ij}) && \text{substitute } \alpha_j \end{aligned}$$

- OLS of this equation gives $\hat{\beta} = 2.1912$, exactly the FE slope (FWL: $X_{ij}$ net of $\bar{X}_j$ is $X_{ij} - \bar{X}_j$).
- $\hat{\gamma} = 3.53 = 5.72 - 2.19$: student-weighted between slope minus within slope.
- Random effects is this model with $\gamma = 0$. Test $\gamma = 0$ with clustered s.e.: $3.53/0.36$, $\chi^2_1 = 96.8$, $p < 0.001$.
- Valid under heteroskedasticity and any within-school correlation, needs no variance difference, and extends to several regressors (add each mean).

# Fixed or Random: How to Choose

| if the question is | use |
|---|---|
| the effect of a member-level variable, and the group effect may be confounded | fixed effects (or Mundlak: same $\hat{\beta}$) |
| the effect of a group-level variable (Catholic) | Mundlak with RE: $\gamma$ protects $\beta$, but Catholic's coefficient still needs $E[a_j \mid \text{Catholic}] = 0$ |
| how much variation lies between groups (Week 4's ICC) | random effects |

- Run Mundlak's test; report it with clustered s.e.s. Rejection says RE's member-level coefficient is biased.
- Not rejecting does not make FE wrong: FE stays consistent either way.
- Either way, FE is not automatically causal: it removes only what group members share.

# From Schools to Years

A panel is the same structure with the person as the group and the years of a life as the members:

| | schools | panel |
|---|---|---|
| group, member | school $j$, student $i$ | person $i$, year $t$ |
| group effect | $\alpha_j$ | $\alpha_i$ |
| within comparison | schoolmates | the same person in other years |

Everything above carries over. Time adds what schoolmates do not have:

- common shocks to all units in a year ($\lambda_t$, two-way fixed effects);
- order: errors correlated over years, past treatment affecting today's outcome, past outcomes affecting today's treatment;
- lagged outcomes as regressors (Nickell bias).

# Panel Data

# Panel Fixed Effects: Goal and Plan

**Goal**: the standard panel toolkit. What each estimator assumes, how to check it, and what to do when the outcome depends on its own past.

1. The model and its three assumptions; two-way fixed effects.
2. Strict exogeneity: why demeaning needs it, when it is causal, a test.
3. First differences, and when FD beats FE.
4. Inference: cluster by unit.
5. Random effects, Hausman, Mundlak on the marriage data; what FE averages.
6. Dynamic panels: Nickell bias, Anderson–Hsiao, Arellano–Bond.
7. Random trends and the impact function; two studies.

Data: NLSY men 1980–87, 545 men × 8 years (`wooldridge::wagepan`) (figs/w10_panel.R, figs/w10_panel_std.R).

# The Error Components Model ★

Unit $i = 1, \ldots, N$, period $t = 1, \ldots, T$; $N$ large, $T$ small:

$$Y_{it} = X_{it}'\beta + \underbrace{\alpha_i}_{\text{unit effect}} + \underbrace{\lambda_t}_{\text{time effect}} + \epsilon_{it}.$$

| | assumption | needed for |
|---|---|---|
| FE.1 | strict exogeneity: $E[\epsilon_{it} \mid X_{i1}, \ldots, X_{iT}, \alpha_i] = 0$ for all $t$ | consistency |
| FE.2 | every column of $X_{it}$ varies over time within some units | identification |
| FE.3 | $\text{Var}(\epsilon_{it} \mid X_i, \alpha_i) = \sigma^2$, no correlation across $t$ | efficiency, usual s.e. |

- Nothing is assumed about $\alpha_i$: it may correlate with $X_{it}$ in any way.
- Random effects adds $E[\alpha_i \mid X_{i1}, \ldots, X_{iT}] = 0$, as with schools.
- FE.3 is usually false; drop it and cluster (below).

Labels as in Wooldridge (2010, ch. 10).

# Two-Way Fixed Effects: Double Demeaning ★

In a balanced panel, average the model over $t$, over $i$, and over both:

$$
\begin{aligned}
Y_{it} &= X_{it}'\beta + \alpha_i + \lambda_t + \epsilon_{it} && \text{the model} \\
\bar{Y}_{i\cdot} &= \bar{X}_{i\cdot}'\beta + \alpha_i + \bar{\lambda} + \bar{\epsilon}_{i\cdot} && \text{average over } t \\
\bar{Y}_{\cdot t} &= \bar{X}_{\cdot t}'\beta + \bar{\alpha} + \lambda_t + \bar{\epsilon}_{\cdot t} && \text{average over } i \\
\bar{Y} &= \bar{X}'\beta + \bar{\alpha} + \bar{\lambda} + \bar{\epsilon} && \text{average over both} \\
Y_{it} - \bar{Y}_{i\cdot} - \bar{Y}_{\cdot t} + \bar{Y} &= (X_{it} - \bar{X}_{i\cdot} - \bar{X}_{\cdot t} + \bar{X})'\beta + \ddot{\epsilon}_{it} && \text{line } 1-2-3+4\text{: } \alpha_i \text{ and } \lambda_t \text{ cancel}
\end{aligned}
$$

- OLS on double-demeaned data = OLS with man and year dummies (FWL). Marriage data: 0.04668 both ways.
- Unbalanced panels have no closed form; software demeans by alternating between unit and period means until nothing changes (`fixest`).

# Strict Exogeneity: What Demeaning Requires ★

With one regressor and unit effects only, the within estimator is

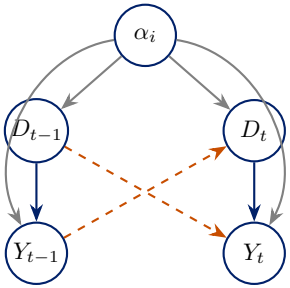
$$\hat{\beta}_{FE} - \beta = \frac{\sum_i \sum_t (X_{it} - \bar{X}_{i\cdot})(\epsilon_{it} - \bar{\epsilon}_{i\cdot})}{\sum_i \sum_t (X_{it} - \bar{X}_{i\cdot})^2} \qquad \text{substitute the demeaned model}$$

$$\bar{\epsilon}_{i\cdot} = \tfrac{1}{T}(\epsilon_{i1} + \cdots + \epsilon_{iT}) \qquad \text{contains every period's error}$$

The numerator has mean zero only if $X_{is}$ is uncorrelated with $\epsilon_{it}$ for *every* pair $s, t$: past, present, and future. That is FE.1.

- $E[\epsilon_{it} \mid X_{it}] = 0$ alone is not enough.
- It fails under **feedback**: a wage shock this year ($\epsilon_{it}$) makes marriage next year ($X_{i,t+1}$) more likely.
- If only feedback fails, FE's bias shrinks like $1/T$; FD's does not (Wooldridge 2010, ch. 10).

# When Is Fixed Effects Causal? (Imai and Kim 2019)




Solid: allowed. Dashed orange: ruled out.

FE removes $\alpha_i$ and nothing else. It needs:

- **No time-varying confounders** of $D_t$ and $Y_t$.
- **No carryover**: past treatment does not affect today's outcome ($D_{t-1} \not\to Y_t$).
- **No feedback**: past outcomes do not affect today's treatment ($Y_{t-1} \not\to D_t$).

Together: strict exogeneity. A man whose wage rises and who then marries is feedback; a marriage effect that builds over years is carryover.

Relaxing these turns FE into matching on treatment histories (Imai, Kim, and Wang 2023; `PanelMatch`) or into an event study (below).

# Testing Strict Exogeneity: Add the Lead

Under FE.1, next year's treatment cannot predict this year's outcome once $\alpha_i$ and this year's treatment are held fixed:

$$Y_{it} = \beta D_{it} + \delta D_{i,t+1} + X_{it}'\gamma + \alpha_i + \lambda_t + \epsilon_{it}, \qquad H_0 : \delta = 0.$$

| marriage data, 1980–86 (lead needs 1987) | estimate | (se) |
|---|---|---|
| married this year | 0.048 | (0.025) |
| married next year | −0.0005 | (0.022) |

- No sign that men marry after a wage jump: $p = 0.98$.
- The s.e. (0.022) is as large as the effect: a small feedback would go unseen.
- This is the first pre-period coefficient of an event study (next section).

Man and year effects, union, experience$^2$; clustered by man (figs/w10_panel_std.R).

# First Differences

Subtract last year's equation from this year's:

$$Y_{it} - Y_{i,t-1} = (X_{it} - X_{i,t-1})'\beta + (\lambda_t - \lambda_{t-1}) + (\epsilon_{it} - \epsilon_{i,t-1}) \quad \text{difference; } \alpha_i - \alpha_i = 0$$

$$\Delta Y_{it} = \Delta X_{it}'\beta + \Delta\lambda_t + \Delta\epsilon_{it} \quad \Delta \text{ for the change}$$

- Consistent if $E[\Delta X_{it}\, \Delta\epsilon_{it}] = 0$: only adjacent periods matter, slightly weaker than FE.1.
- $T = 2$: FD and FE are the same estimator. Marriage data, 1986–87 only: 0.00488 both ways.
- $T > 2$: they differ. Marriage data, all 8 years: FE 0.047 (0.021), FD 0.038 (0.024).
- Both consistent under FE.1. Which is more precise depends on how $\epsilon_{it}$ is correlated over time.

# FE or FD? It Depends on the Serial Correlation

If FE.3 holds ($\epsilon_{it}$ uncorrelated over time, variance $\sigma^2$):

$$\text{Cov}(\Delta\epsilon_{it}, \Delta\epsilon_{i,t-1}) = \text{Cov}(\epsilon_{it} - \epsilon_{i,t-1},\ \epsilon_{i,t-1} - \epsilon_{i,t-2}) = -\sigma^2$$ only $\epsilon_{i,t-1}$ is shared

$$\text{Corr}(\Delta\epsilon_{it}, \Delta\epsilon_{i,t-1}) = -\sigma^2/2\sigma^2 = -0.5$$ $\text{Var}(\Delta\epsilon_{it}) = 2\sigma^2$

| $\epsilon_{it}$ is | $\Delta\epsilon_{it}$ is | efficient |
|---|---|---|
| uncorrelated over time | correlated, $-0.5$ | FE |
| a random walk | uncorrelated, 0 | FD |

**Check**: regress the FD residual on its own lag. Marriage data: $-0.40$ (0.02). Neither $-0.5$ nor 0: errors are positively correlated over time, short of a random walk. Neither estimator is fully efficient; cluster either way.

# Inference: Cluster by Unit ★

Errors of the same man are correlated across years (previous frame: FE.3 fails). The cluster-robust variance allows any correlation within a unit (Arellano 1987):

$$\widehat{\mathrm{Var}}(\hat{\beta}_{FE}) = \Big(\sum_i \ddot{\boldsymbol{X}}_i'\ddot{\boldsymbol{X}}_i\Big)^{-1}\Big(\sum_i \ddot{\boldsymbol{X}}_i'\hat{\boldsymbol{e}}_i\hat{\boldsymbol{e}}_i'\ddot{\boldsymbol{X}}_i\Big)\Big(\sum_i \ddot{\boldsymbol{X}}_i'\ddot{\boldsymbol{X}}_i\Big)^{-1}$$

$\ddot{\boldsymbol{X}}_i$: man $i$'s $T$ rows of demeaned regressors; $\hat{\boldsymbol{e}}_i$: his $T$ residuals.

- Week 2's sandwich, with a man's $T$ years as one cluster. Valid as $N \to \infty$ with $T$ fixed.
- Marriage data: s.e. 0.0183 assuming FE.3, 0.0210 clustered (15% larger).
- With a treatment that is persistent over time, the gap is much larger: placebo laws in the DiD section.

# Random Effects in a Panel

As with schools, RE is OLS after subtracting a share $\theta$ of each man's mean. In a balanced panel, one $\theta$ for everyone:

$$Y_{it} - \theta \bar{Y}_{i\cdot} = (X_{it} - \theta \bar{X}_{i\cdot})'\beta + \ldots, \qquad \theta = 1 - \sqrt{\frac{\sigma_\epsilon^2}{\sigma_\epsilon^2 + T\sigma_\alpha^2}}$$

$$\hat{\theta} = 1 - \sqrt{\frac{0.123}{0.123 + 8(0.105)}} = 0.64 \qquad \text{marriage data}$$

- The man effect is 46% of the error variance; RE removes 64% of each man's mean.
- RE needs FE.1 *and* $E[\alpha_i \mid X_{i1}, \ldots, X_{iT}] = 0$: the man effect unrelated to his whole marriage history.
- RE's 0.064 lies between pooled 0.108 and FE 0.047 (next frame): it partly uses the between-man comparison, and so can estimate education and race.

Swamy–Arora variance components, `plm` (figs/w10_panel_std.R).

# The Marriage Wage Premium, Four Ways

NLSY men observed every year 1980–87 (545 men, 4,360 person-years; Vella and Verbeek 1998). Log hourly wage on married, union, and year effects; standard errors clustered by man (figs/w10_panel.R).

| estimator | uses | married | (se) |
|---|---|---|---|
| pooled OLS (+ education, race, experience) | all variation | 0.108 | (0.026) |
| random effects | a GLS mix | 0.064 | (0.017) |
| fixed effects | within-man change | 0.047 | (0.021) |
| first differences | year-to-year change | 0.038 | (0.024) |

- The premium falls by more than half once each man is compared with himself: men with higher earning potential are more likely to marry.
- Hausman test of FE against RE: $\chi^2_{10} = 37.0$, $p < 0.001$.
- 310 of 545 men change marital status; the other 235 contribute nothing to the FE estimate of the marriage effect (next frames).

# Fixed Effects as a Regression on Unit Means (Mundlak 1978)

Suppose the unit effect depends on the regressors only through their mean, $\alpha_i = \gamma \bar{D}_i + a_i$ with $a_i$ unrelated to $D_{i1}, \ldots, D_{iT}$:

$$
\begin{aligned}
Y_{it} &= \beta D_{it} + \alpha_i + \epsilon_{it} && \text{the FE model} \\
&= \beta D_{it} + \gamma \bar{D}_i + (a_i + \epsilon_{it}) && \text{substitute } \alpha_i \\
&= \beta (D_{it} - \bar{D}_i) + (\beta + \gamma) \bar{D}_i + (a_i + \epsilon_{it}) && \text{regroup: within and between parts}
\end{aligned}
$$

- Pooled OLS of this equation gives exactly the FE $\hat{\beta}$ (FWL: $D_{it}$ net of $\bar{D}_i$ is $D_{it} - \bar{D}_i$). Marriage data: 0.0467 both ways.
- $\gamma$ is the between-minus-within gap: 0.094 (0.045) here. Random effects is this model with $\gamma = 0$; testing $\gamma = 0$ is a Hausman test that works with clustered errors.
- Sociology's **hybrid model** (Allison 2009) is the last line plus time-invariant covariates, which FE cannot estimate; their coefficients are not causal merely because the model is FE-consistent.

# What Fixed Effects Averages

Binary $D$, unit $i$ treated in $m_i$ of $T$ periods, $p_i = m_i/T$. With tildes for deviations from unit means:

$$\hat{\beta}_{FE} = \frac{\sum_i \sum_t \tilde{D}_{it}\tilde{Y}_{it}}{\sum_i \sum_t \tilde{D}_{it}^2} = \sum_i \underbrace{\frac{\sum_t \tilde{D}_{it}^2}{\sum_j \sum_t \tilde{D}_{jt}^2}}_{w_i} \underbrace{\frac{\sum_t \tilde{D}_{it}\tilde{Y}_{it}}{\sum_t \tilde{D}_{it}^2}}_{b_i}$$

multiply and divide by $\sum_t \tilde{D}_{it}^2$

$$\sum_t \tilde{D}_{it}^2 = T\, p_i(1-p_i)$$

variance of a 0/1 variable

$b_i$ is unit $i$'s own before–after difference. Units are weighted by how much their treatment varies, not by their number.

| years married, of 8 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|
| weight per man, $8p(1-p)$ | 0.88 | 1.50 | 1.88 | 2.00 | 1.88 | 1.50 | 0.88 |
| share of total weight | .08 | .11 | .19 | .15 | .22 | .14 | .12 |

Man effects only: $\hat{\beta}_{FE} = 0.243$ = weighted mean of 310 men's $b_i$ (unweighted 0.255). Adding year effects: 0.061; wages rise with age for everyone, and marriage comes with age (figs/w10_panel.R).

# What Fixed Effects Costs ★

- **Precision**. You are using only within-unit variation. If $D_{it}$ barely moves within units, the standard error explodes, FWL again.
- **Measurement error is amplified**. Differencing removes signal and keeps noise, so attenuation bias gets worse.
- **Nickell bias**: adding a lagged outcome to a fixed-effects model creates its own bias (dynamic panels, below).
- Always cluster on the unit (Week 2). Serial correlation within units is the norm.

# Dynamic Panels: Nickell Bias

The outcome depends on its own past:

$$Y_{it} = \rho Y_{i,t-1} + X_{it}'\beta + \alpha_i + \epsilon_{it}.$$

FE.1 fails by construction: $Y_{i,t-1}$ contains $\epsilon_{i,t-1}$, which is in $\bar{\epsilon}_{i\cdot}$. Demeaned lag and demeaned error are correlated (Nickell 1981):

$$\text{plim}_{N\to\infty}\ \hat{\rho}_{FE} - \rho \ \approx\ -\frac{1+\rho}{T-1}$$

large-$T$ approximation; exact formula in the script

| simulation, $\rho = 0.5$, $N = 500$, 200 draws | $T = 4$ | $T = 8$ | $T = 16$ |
|---|---|---|---|
| mean of $\hat{\rho}_{FE}$ | $-0.04$ | 0.27 | 0.40 |
| Nickell's exact fixed-$T$ value | $-0.04$ | 0.27 | 0.39 |

Large $N$ does not help; only large $T$ does (figs/w10_panel_std.R).

# Anderson–Hsiao: Difference, Then Instrument

First-difference to remove $\alpha_i$:

$$\Delta Y_{it} = \rho\,\Delta Y_{i,t-1} + \Delta X_{it}'\beta + \Delta\epsilon_{it} \qquad \alpha_i \text{ gone}$$

$$\Delta Y_{i,t-1} = Y_{i,t-1} - Y_{i,t-2}, \quad \Delta\epsilon_{it} = \epsilon_{it} - \epsilon_{i,t-1} \qquad \text{both contain } \epsilon_{i,t-1}\text{: endogenous}$$

Instrument $\Delta Y_{i,t-1}$ with $Y_{i,t-2}$ (Anderson and Hsiao 1981), the Week 7 conditions:

- **Relevance**: $Y_{i,t-2}$ is part of $\Delta Y_{i,t-1}$.
- **Exogeneity**: $Y_{i,t-2}$ depends on errors up to $t-2$; $\Delta\epsilon_{it}$ on $t$ and $t-1$. Holds only if $\epsilon_{it}$ is serially uncorrelated.

| same simulation, $\rho = 0.5$ | $T = 4$ | $T = 8$ | $T = 16$ |
|---|---|---|---|
| mean of $\hat{\rho}_{AH}$ (sd) | 0.51 (0.13) | 0.51 (0.05) | 0.50 (0.03) |

# Arellano–Bond: Use Every Valid Lag

If $\epsilon_{it}$ is serially uncorrelated, *every* lag from $t-2$ back is a valid instrument for period $t$'s differenced equation:

$$E\big[Y_{i,t-s}\,\Delta\epsilon_{it}\big] = 0, \qquad s \geq 2,\ t = 3, \ldots, T.$$

GMM combines all these moment conditions (Arellano and Bond 1991).

- More instruments, more precision: period $t$ has $t-2$ of them. With $T = 8$ and year effects: 29 instruments.
- **The key check**: the differenced residuals $\Delta\hat{\epsilon}_{it}$. First-order correlation is expected ($-0.5$ under no serial correlation, two frames back). Second-order correlation must be zero; if it is not, $\epsilon_{it}$ is serially correlated and $Y_{i,t-2}$ is invalid. Start at $Y_{i,t-3}$.
- Too many instruments overfit the first stage and pull GMM toward FE-style bias; report the count (Roodman 2009).

# A Dynamic Wage Equation on the Marriage Data

log wage on its lag, married, union; year effects; robust s.e.; `fixest` (AH), `plm::pgmm` one-step (AB) (figs/w10_panel_std.R):

| estimator | lagged wage $\hat{\rho}$ | married | note |
|---|---|---|---|
| pooled OLS | 0.60 (0.03) | 0.056 (0.015) | $\alpha_i$ in the error: $\hat{\rho}$ too high |
| FE | 0.06 (0.02) | 0.062 (0.021) | Nickell: $\hat{\rho}$ too low |
| Anderson–Hsiao, $Y_{t-2}$ | 0.12 (0.03) | 0.046 (0.022) | first-stage $F = 830$ |
| Arellano–Bond, $Y_{t-2}$ on | 0.16 (0.04) | 0.044 (0.023) | 2nd-order test $z = 2.18$, $p = 0.03$ |
| Arellano–Bond, $Y_{t-3}$ on | 0.68 (0.14) | 0.031 (0.031) | 2nd-order test $p = 0.01$ |
| Anderson–Hsiao, $Y_{t-3}$ | 0.81 (0.27) | 0.027 (0.034) | first-stage $F = 12$ |

- The second-order test rejects: wage shocks persist, so $Y_{t-2}$ is not a valid instrument, and the rows using it are biased.
- Lag 3 is weak ($F = 12$) and the test still rejects: $\hat{\rho}$ jumps to 0.7–0.8 with large s.e. Persistence is not pinned down in eight years.
- The marriage coefficient, 0.03–0.06, is far less sensitive than $\hat{\rho}$.

# Fixed Effects or a Lagged Outcome?

Sociologists routinely control for last year's outcome, to absorb differences between units that are *changing* rather than fixed. It is a reasonable instinct. But combining that move with unit fixed effects creates a new bias, and it is worth knowing why before you reach for it.

- **FE and LDV embody different selection stories**: FE assumes selection on time-invariant traits; LDV assumes selection on prior outcomes. They can both be valid, but they are different assumptions and will generally give different answers.
- Angrist and Pischke (2009) suggest running both: under some conditions the two bracket the truth, so agreement is reassuring and disagreement tells you how much the selection story matters. Treat that as a heuristic, not a theorem.

# Random Trends: Each Man His Own Slope

FE allows men to differ in wage *levels*. If men on steeper wage paths marry, FE still mistakes their growth for a marriage effect.

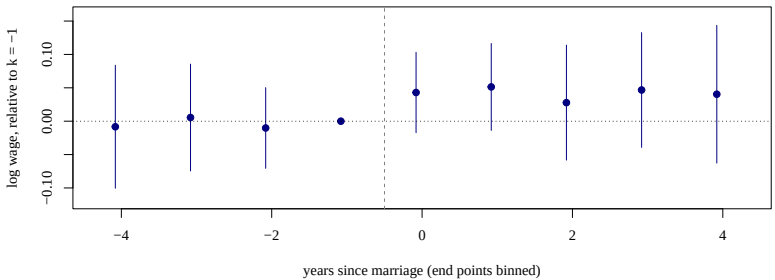
$Y_{it} = \alpha_i + \gamma_i\, x_{it} + \beta D_{it} + \epsilon_{it}$ own intercept $\alpha_i$, own slope $\gamma_i$ in experience $x$

$\ddot{Y}_{it} = \beta \ddot{D}_{it} + \ddot{\epsilon}_{it}$ double dots: residuals from man $i$'s own regression on $(1, x_{it})$

- The **random trend model** (Wooldridge 2010, ch. 11); sociology calls it **FEIS**, fixed effects with individual slopes: detrend each unit separately, then pool (Brüderl and Ludwig 2015; Rüttenauer and Ludwig 2023). Needs at least three observations per unit.
- Identifies $\beta$ from *departures* from each man's own trajectory at marriage: a jump, not a steeper path.
- Our 8-year panel: FE 0.047 (0.021), FEIS 0.055 (0.024). Eight years per man is short for estimating a slope; the NLSY79 over 34 years (next frames) tells a different story.

# The Impact Function: an Event Study Around Marriage

Men who marry once in 1981–87 (239) and men never married in the panel (162); event-time dummies, man and year effects, $k = -1$ omitted (figs/w10_panel.R):




- Before marriage: −0.008, 0.006, −0.010. After: 0.04–0.05, each within noise.
- The impact function shows what one coefficient hides: anticipation, build-up, fade-out (Ludwig and Brüderl 2021). It is a DiD event study with marriage as the event (next section).

# In the Literature: Budig and England (2001)

**The question.** Why do mothers earn less? NLSY79 women 1982–93: 5,287 women, 41,842 person-years; log hourly wage on number of children.

| penalty per child, their Table 2 | fixed effects | pooled OLS |
|---|---|---|
| no controls | −0.068 (0.004) | −0.081 (0.002) |
| + marital status | −0.073 | −0.081 |
| + experience, seniority, education | −0.047 | −0.018 |
| + job characteristics | −0.037 | −0.012 |

- About 7% per child within women, close to OLS: little selection of lower-wage women into motherhood on fixed traits.
- Experience (lost years at work) explains about a third of the FE penalty; the rest remains. Note that experience is *post*-birth, a mediator: controlling for it answers a different question.

# In the Literature: Ludwig and Brüderl (2018)

**The question.** Is the male marriage wage premium causal, or do men on steeper careers marry? A test of specialization theory.

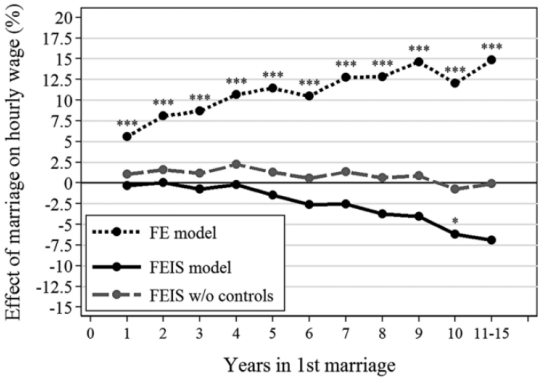
**The data.** NLSY79, 1979–2012: 4,287 employed men, 49,801 person-years; men observed never married at least once.

**The design.** Four estimators of one coefficient:

| model | allows men to differ in | marriage premium |
|---|---|---|
| pooled OLS | nothing | 17.7% |
| FE | wage level | 8.3% |
| FE, group slopes | level; growth by future marital status | 4.3% |
| FEIS | level and own wage growth | 0.5% (n.s.) |

Their Figure 3. Before marrying, future husbands' wages grew 5% per year of experience; never-married men's, 3%.

# Ludwig and Brüderl: the Premium Over Time




Their Figure 4: effect by years since first marriage.

- FE: the premium grows to 11.4% after 5 years and 14.8% after 11–15, the pattern read as evidence of specialization.
- FEIS: about zero throughout.
- The growing "premium" is the steeper wage path the men were already on: selection on growth, not an effect of marriage.
- Generalizes: an FE impact function that keeps rising after the event is a warning sign of selection on slopes.

# DID

# The Canonical $2 \times 2$ ★

Two groups, two periods. One group is treated in period 2.

| | Before | After | Difference |
|---|---|---|---|
| Treated | $E[Y \mid g{=}1, t{=}1]$ | $E[Y \mid g{=}1, t{=}2]$ | $\Delta_1$ |
| Control | $E[Y \mid g{=}0, t{=}1]$ | $E[Y \mid g{=}0, t{=}2]$ | $\Delta_0$ |
| | | | $\tau = \Delta_1 - \Delta_0$ |

- The first difference removes anything fixed about each group.
- The second difference removes anything common to both groups over time.
- Equivalent regression: $Y_{it} = \alpha + \gamma\,\text{treated}_i + \delta\,\text{post}_t + \tau\,(\text{treated}_i \times \text{post}_t) + \epsilon_{it}$.

# The $2 \times 2$ in the Literature: Card and Krueger (1994)

New Jersey raised its minimum wage from \$4.25 to \$5.05 in April 1992; Pennsylvania did not. 410 fast-food restaurants surveyed in February and November 1992.

| FTE employment per store (se), their Table 3 | PA (79) | NJ (331) | NJ − PA |
|---|---|---|---|
| February (before) | 23.33 (1.35) | 20.44 (0.51) | −2.89 (1.44) |
| November (after) | 21.17 (0.94) | 21.03 (0.52) | −0.14 (1.07) |
| change | −2.16 (1.25) | 0.59 (0.54) | 2.76 (1.36) |

- $2.76 = 0.59 - (-2.16)$: Pennsylvania's decline stands in for what New Jersey would have done without the law.
- Parallel trends here is one claim about one region in one season. Their within-NJ check: high-wage NJ stores, not bound by the law, changed like PA (−2.16 vs. −2.28, balanced sample).

# The $2 \times 2$ in Regression Form

Write $g_i \in \{0, 1\}$ for group and $t \in \{1, 2\}$ for period:

$$Y_{it} = \alpha + \gamma\, g_i + \delta\, \text{post}_t + \tau\, (g_i \times \text{post}_t) + \epsilon_{it}.$$

Read off the four cell means:

| | Before | After | Difference |
|---|---|---|---|
| Treated ($g = 1$) | $\alpha + \gamma$ | $\alpha + \gamma + \delta + \tau$ | $\delta + \tau$ |
| Control ($g = 0$) | $\alpha$ | $\alpha + \delta$ | $\delta$ |
| | | | $\tau$ |

- $\gamma$ absorbs fixed differences in level; $\delta$ absorbs the common time trend; $\tau$ is what is left.

# The $2 \times 2$ in Regression Form (cont.)

- With unit and period fixed effects instead of $g_i$ and $\text{post}_t$; this is the two-way fixed-effects model, identical when there are two groups and two periods, and, as we will see, not identical otherwise.

# Parallel Trends ★

## The identifying assumption

$$E[Y_{i2}(0) - Y_{i1}(0) \mid g_i = 1] = E[Y_{i2}(0) - Y_{i1}(0) \mid g_i = 0].$$

*Absent treatment*, the two groups' average outcomes would have moved in parallel.

- Levels may differ arbitrarily. Only the *trends* in the untreated potential outcome must agree.
- Untestable: it concerns a counterfactual trend. Pre-treatment parallelism is suggestive evidence, not proof.
- **Not scale-free**: parallel trends in levels and parallel trends in logs are different assumptions, and both cannot generally hold. The next slide shows why this is a real choice and not a technicality.

# Parallel Trends ★ (cont.)

- Needs **no anticipation**: units must not change behaviour before treatment in expectation of it.

# What the Double Difference Identifies, Exactly ★

Decompose the treated group's observed change. Its period-2 outcome is $Y_{i2}(1)$; add and subtract the counterfactual $Y_{i2}(0)$:

$$\Delta_1 = E[Y_{i2}(1) - Y_{i1}(0) \mid g_i{=}1] = \underbrace{E[Y_{i2}(1) - Y_{i2}(0) \mid g_i{=}1]}_{\text{ATT}} + \underbrace{E[Y_{i2}(0) - Y_{i1}(0) \mid g_i{=}1]}_{\text{treated group's } \textit{untreated} \text{ trend}}.$$

The control group supplies an estimate of that second, unobservable term: $\Delta_0 = E[Y_{i2}(0) - Y_{i1}(0) \mid g_i{=}0]$. Subtracting,

$$\boxed{\Delta_1 - \Delta_0 \;=\; \text{ATT} \;+\; \underbrace{E[\Delta Y(0) \mid g{=}1] - E[\Delta Y(0) \mid g{=}0]}_{\text{differential trend; } = 0 \text{ under parallel trends}}}$$

- The estimand is the ATT, not the ATE: we borrowed a counterfactual trend *for the treated group*. Nothing in the design identifies what treatment would have done to the controls.

# What the Double Difference Identifies, Exactly ★ (cont.)

- The bias term *is* the parallel-trends violation, exactly. Every diagnostic in this lecture (pre-trends, placebos, Rambachan–Roth bounds) is aimed at that one object.
- Note also what was *not* assumed: no restriction on the level difference $\gamma$, and no randomization anywhere.

# Parallel Trends Is Not Scale-Free ★

Suppose untreated outcomes would have grown by 10% in both groups, from a base of 20 in the treated group and 40 in the control group.

| | Before | After | Change |
|---|---|---|---|
| Treated | 20 | 22 | +2 |
| Control | 40 | 44 | +4 |

- Parallel trends holds in logs (both +10%) but fails in levels (+2 versus +4). A DID in levels would report a spurious −2.
- The two assumptions are mutually exclusive unless base levels are equal. There is no neutral choice, and the transformation is part of the identifying assumption, not a presentational detail.

# Parallel Trends Is Not Scale-Free ★ (cont.)

- Choose on substantive grounds, is the intervention plausibly additive or multiplicative?, and *before* looking at the results.
- One diagnostic: if pre-treatment levels are far apart, be especially careful. If they are similar, the choice matters less.

# When Does the Scale Not Matter? (Roth and Sant'Anna 2023)

Parallel trends in levels, in logs, in ranks: when can all hold at once?

**Result**

Parallel trends holds for *every* monotone transformation of $Y$ if and only if the whole *distribution* of $Y(0)$ satisfies a parallel-trends condition. Essentially only two cases deliver that: treatment is as good as random, or $Y(0)$'s distribution does not change over time for the groups that differ.

- In any real DiD the scale is therefore part of the assumption. Choose it before seeing results, on substantive grounds (previous frame).
- The distributional condition is testable in the pre-period: compare the groups' outcome distributions, not only means.
- For effects on whole distributions under weaker assumptions: Callaway and Li (2019) (later frame).

# Parallel Trends Given Covariates: Three Estimators

Assume parallel trends only within levels of pre-treatment $X$; two periods, $\Delta Y_i = Y_{i2} - Y_{i1}$, $e(x) = \Pr(D_i = 1 \mid X_i = x)$, $m_{0,\Delta}(x) = E[\Delta Y_i \mid X_i = x, D_i = 0]$:

$$\text{regression:} \quad \text{ATT} = E[\Delta Y_i \mid D_i = 1] - E[m_{0,\Delta}(X_i) \mid D_i = 1]$$

$$\text{weighting:} \quad \text{ATT} = E\Big[\frac{D_i - e(X_i)}{\Pr(D=1)\,(1 - e(X_i))}\,\Delta Y_i\Big]$$

$$\text{doubly robust:} \quad \text{ATT} = E\big[(w_1(D_i) - w_0(D_i, X_i))\,(\Delta Y_i - m_{0,\Delta}(X_i))\big]$$

with $w_1 = D/E[D]$ and $w_0 \propto e(X)(1-D)/(1-e(X))$, normalized to mean one (Heckman, Ichimura, and Todd 1997; Abadie 2005; Sant'Anna and Zhao 2020).

- This is Week 6 applied to $\Delta Y$: matching and weighting on the *change*, not the level.
- The DR version is consistent if either $e$ or $m_{0,\Delta}$ is right (`DRDID`); it is what `did` uses inside each $ATT(g,t)$.

# Why Not Just Add Covariates to TWFE?

$Y_{it} = \alpha_i + \lambda_t + \beta D_{it} + X_{it}'\delta + \epsilon_{it}$ is the default. Even with two periods it fails when (Caetano and Callaway 2024):

1. $X_{it}$ is affected by treatment (a "bad control", Week 5);
2. trends depend on the *level* of $X$, not only its change: the within transformation keeps only changes;
3. trends depend on time-invariant $X$ (region, education), which the unit effects absorb;
4. effects vary with $X$, so the regression weights the subgroups in its own way (Week 6's regression weights again);
5. the linear form is wrong.

Fix: use pre-treatment $X$ (or $X$ at a baseline period) in one of the three estimators on the previous frame; never post-treatment $X$.

# Inference: Placebo Laws on Real Data

Bertrand, Duflo, and Mullainathan (2004): give fake laws to random states in CPS wage data, 1979–99. Ordinary standard errors on the micro data reject "no effect" 67.5% of the time at 5%; 44% even after aggregating to a state-year panel.

**Same experiment on our state panel** (female suicides, 51 states, 1964–96): 25 random states get a fake law from a random year, 1,000 times (figs/w10_placebo.R).

| standard error | share of fake laws "significant" at 5% |
|---|---|
| assumes independent state-years | 0.477 |
| clustered by state | 0.049 |
| clustered by state, only 3 treated states | 0.139 |

- Residuals are serially correlated (lag-1 correlation 0.36): 33 years of one state are far fewer than 33 independent observations.
- With few *treated* clusters, clustering also fails; use randomization inference or methods built for it (Conley and Taber 2011; MacKinnon and Webb 2018).

# Event Studies ★

Generalize to many periods by estimating an effect for each period relative to treatment:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \tau_k \cdot \mathbb{1}\{t - g_i = k\} + \epsilon_{it},$$

where $g_i$ is unit $i$'s treatment date and $k = -1$ is the omitted reference.

- **Pre-period coefficients** ($k < 0$) are the parallel-trends check. Flat and near zero is what you want to see.
- **Post-period coefficients** trace out the dynamics: does the effect grow, fade, or appear only after a lag?
- Caution: a flat pre-trend is weak evidence. The test often has low power, and conditioning your analysis on passing it distorts inference (Roth 2022). Report the power of your pre-trend test.

# Event Studies: Specification Details That Matter

That specification looks innocent. Three choices inside it change the picture, and each must be reported.

- **Normalization**: one lead must be omitted (here $k = -1$). Every $\tau_k$ is relative to it, so the picture shifts if you change the reference period. Say which you used.
- **Binning the endpoints**: with unbalanced exposure, far leads and lags are estimated off few units. Bin them ($k \leq -5$, $k \geq 5$) or restrict to a balanced window, and say which.
- **Never-treated units** anchor the time effects $\lambda_t$. If there are none, the $\lambda_t$ are identified only off variation in timing, and the contamination problem below is unavoidable.

# Pre-Trend Tests Have Low Power (Roth 2022)

**The practice**: estimate the leads, see that they are insignificant, proceed as if parallel trends holds.

**The audit**: 12 event-study papers in leading economics journals (2014–18) with replication data. Roth asks what happens under a linear violation of parallel trends that the usual pre-test would catch only half the time.

- The bias such a violation produces is often as large as the estimated effect, sometimes larger.
- In the worst case a nominal 95% interval covers the truth 24% of the time.
- **Conditioning on passing makes it worse**: among draws that pass the pre-test, the bias is usually larger, sometimes twice as large, because passing selects draws whose noise happened to offset the trend.

Practical: report the power of the pre-test against violations you care about (`pretrends`), and replace "test and proceed" with sensitivity analysis (Rambachan and Roth 2023, below).

# Base Periods: Why Event-Study Plots Disagree (Roth 2026)

No effect at all; untreated outcomes of the treated group drift by $\gamma$ per period relative to controls; treatment starts at $r = 0$.

| | | |
|---|---|---|
| universal base ($r = -1$): | $\hat{\beta}_r = \gamma\,(r+1)$ for every $r$ | as TWFE: one straight line |
| varying base, $r < 0$: | $\hat{\beta}_r = \gamma$ | compares $r$ with $r-1$ |
| varying base, $r \geq 0$: | $\hat{\beta}_r = \gamma\,(r+1)$ | compares $r$ with $-1$ |

- Callaway–Sant'Anna's default ("varying") draws a flat pre-period and a rising post-period: a kink at zero that looks like an effect. The imputation estimator's default draws a jump.
- The pre-period points answer different questions under the two bases. Do not read a varying-base plot with TWFE eyes, and do not feed it to `HonestDiD`.
- For plots: `base_period = "universal"` in `did`. All event studies in this lecture use it.

# Drawing an Event Study (Freyaldenhoven, Hansen, Pérez Pérez, and Shapiro 2025)

- **Normalize** $\hat{\beta}_{-1} = 0$ and label that point with the outcome's mean, so readers can judge magnitudes.
- **Two kinds of bands**: pointwise 95% intervals, and a sup-$t$ band that covers the whole path at once. A path shape ("no pre-trend at all leads") is a joint claim; only the uniform band tests it.
- **Report two $p$-values** in the legend: no pre-trends (leads jointly zero) and effects levelling off (last lags equal).
- **Show the least wiggly confound** consistent with the leads: if a smooth trend through the pre-period would explain the post-period, the plot should show it.
- **Bin or trim the end points** and say which.

Freyaldenhoven, Hansen, and Shapiro (2019): a covariate affected by the confound but not by the policy can identify the effect even with pre-trends (instrument the confound with leads of the policy).

# In the Literature: Kleven, Landais, and Søgaard (2019)

**The question.** How much of the gender gap in earnings is caused by children? Danish registers 1980–2013; parents with a first birth 1985–2003, followed from 5 years before to 10 years after.

**The design**: an event study around the first birth, separately for mothers and fathers, with age and year dummies:

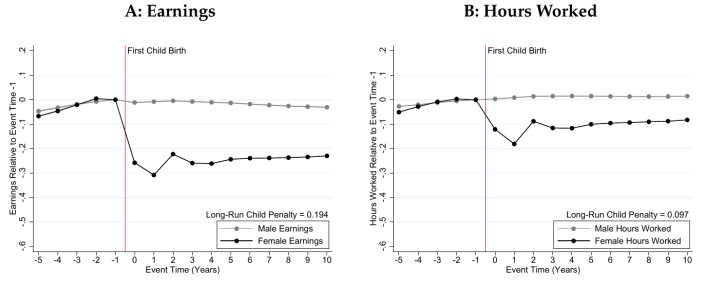
$$Y_{ist}^g = \sum_{j \neq -1} \alpha_j^g \mathbb{1}\{j = t\} + \sum_k \beta_k^g \mathbb{1}\{k = \text{age}_{is}\} + \sum_y \gamma_y^g \mathbb{1}\{y = s\} + \nu_{ist}^g$$

$$P_t = (\hat{\alpha}_t^m - \hat{\alpha}_t^w)/E[\tilde{Y}_{ist}^w \mid t]$$

with $t$ event time, $s$ year, and $\tilde{Y}$ the prediction from age and year alone. $P_t$ is the **child penalty**: the gap that opens, as a share of women's counterfactual earnings.

- No control group of non-parents: identification is smoothness, the same argument as RD in time, with fathers as a check.

# Kleven, Landais, and Søgaard: Results




Their Figure 1, panels A–B (accepted manuscript, 2018).

- Mothers' earnings fall almost 30% at birth and settle about 20% below the counterfactual; fathers' do not move. Long-run penalty: earnings 0.194, hours 0.097, participation 0.130, wage rate 0.091.
- The share of gender earnings inequality due to children rose from about 40% (1980) to about 80% (2013).

# Staggered

# The Setting Everyone Uses ★

Units adopt the policy at *different times*: states pass a law in 1995, 2001, 2008. The standard specification is

$$Y_{it} = \alpha_i + \lambda_t + \beta\, D_{it} + \epsilon_{it}, \qquad D_{it} = \mathbb{1}\{t \geq g_i\}.$$

For thirty years this was assumed to estimate "the" treatment effect. It does when the effect is constant across units and over time since treatment , but with heterogeneous or dynamic effects it need not estimate any sensible average at all.

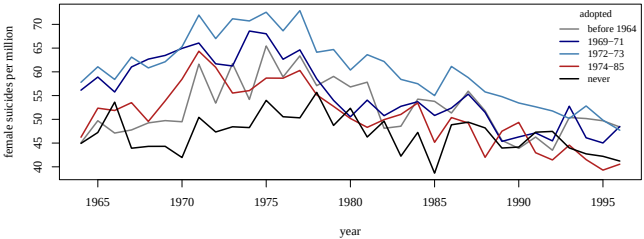
**Goodman-Bacon (2021)**

The TWFE estimate is a weighted average of *all possible* $2 \times 2$ DID comparisons in the data, including comparisons that use already-treated units as controls.

# Running Example: Unilateral Divorce and Female Suicide

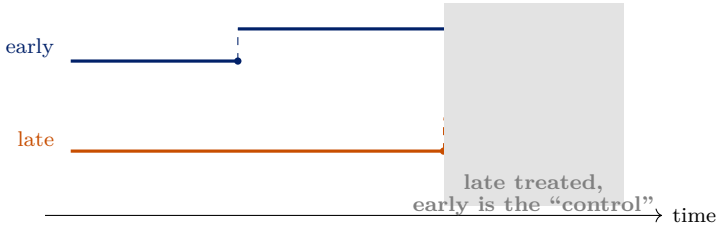
Stevenson and Wolfers (2006): did letting either spouse end a marriage reduce women's suicide? States adopted unilateral divorce in different years.

- 51 states, 1964–96; outcome: female suicides per million women (mean 51 in 1964).
- 9 states adopted before 1964 (always treated), 37 adopted 1969–85, 5 never.




Mean rate by adoption cohort. Their event study (percent change): −9.7% (2.3) averaged over 20 years, −18.7% (3.2) after 19+ years; nothing for men. Data: `bacondecomp::divorce`, crude rates (figs/w10_divorce.R).

# The Forbidden Comparison ★




- In the shaded window, TWFE uses the *early-treated* unit as a control for the late-treated one.
- If the early unit's effect is still evolving, that evolution is subtracted from the late unit's effect, and enters with a negative weight.
- With dynamic effects, TWFE can be the wrong *sign* even when every unit-level effect is positive.

# The Forbidden Comparison, in Algebra ★

Take the late-versus-early $2 \times 2$ from the shaded window: late switches from untreated to treated, early is *treated throughout*. Suppose parallel trends holds exactly and untreated outcomes are flat, so every change in $Y$ is a treatment effect. Then

$$\hat{\beta}^{2\times 2}_{\text{late vs. early}} = \underbrace{\text{ATT}_{\text{late}}(\text{post})}_{\text{what we want}} - \underbrace{\Big[\text{ATT}_{\text{early}}(\text{post}) - \text{ATT}_{\text{early}}(\text{pre})\Big]}_{\text{the } \textit{growth} \text{ in early's effect over the window}}.$$

With numbers; no noise, no trends, every true effect positive:

| | $t = 3$ | $t = 4$ | Change |
|---|---|---|---|
| Late (treated at $t$=4; effect 1) | 0 | 1 | +1 |
| Early (treated at $t$=2; effects 1, 2, 3) | 2 | 3 | +1 |
| | | | $\hat{\beta}^{2\times 2} = 0$ |

# The Forbidden Comparison, in Algebra ★ (cont.)

- Late's true effect is $+1$; this comparison returns 0. If early's effect grew by 2 per period instead, it would return $-1$.
- If effects were constant $(1, 1, 1)$, the bracketed term would vanish and the comparison would be harmless, which is exactly why the problem stayed invisible for thirty years of constant-effect thinking.

# The Goodman-Bacon Decomposition ★

With staggered timing, the TWFE estimate is a weighted average of every possible $2 \times 2$ comparison in the data:

$$\hat{\beta}^{\text{TWFE}} = \sum_{\text{pairs}} s_{ab}\, \hat{\beta}^{2\times 2}_{ab}, \qquad \sum s_{ab} = 1,$$

where the weights $s_{ab}$ depend on group sizes and on the *variance of treatment* within each pair, largest for pairs whose treatment switches mid-window. (The next frame writes them out.)

Three kinds of comparison arise:

1. Treated versus **never-treated**, clean.
2. **Earlier** treated versus **later** treated, before the later group switches, clean.
3. **Later** treated versus **earlier** treated, after the earlier group switches, the forbidden comparison.

Type 3 uses already-treated units as controls. If their effects are still evolving, that evolution is subtracted from the late group's effect.

# The Decomposition on the Divorce Data

TWFE: $\hat{\beta} = -3.05$ (2.32). All 156 $2 \times 2$ comparisons, grouped (figs/w10_divorce.R; Goodman-Bacon's own numbers, age-adjusted rates, 49 states, in grey):

| comparison | weight | average $2 \times 2$ | weight | average |
|---|---|---|---|---|
| timing states vs. always treated (pre-1964) | 0.41 | −6.92 | 0.38 | −7.04 |
| treated vs. never treated | 0.23 | −5.21 | 0.24 | −5.33 |
| earlier treated vs. later (not yet) treated | 0.11 | 1.25 | 0.11 | −0.19 |
| later treated vs. earlier (already) treated | 0.25 | 3.36 | 0.26 | 3.51 |
| TWFE = weighted sum | | −3.05 | | −3.08 |

- A quarter of the weight sits on comparisons that use already-treated states as controls; their average has the *opposite sign*.
- 41% comes from always-treated states as controls: their whole period is post-treatment, so they are clean only if their effects are constant.

# The Goodman-Bacon Weights, Explicitly ★

Let $n_k, n_l, n_U$ be the groups' sample shares and $\bar{D}_j$ the fraction of the panel that group $j$ spends treated ($k$ treated earlier than $l$). Up to a common normalization, the weight on each $2 \times 2$ is the product of the pair's group sizes and the *variance of the treatment dummy within that pair's window*:

$$s_{kU} \propto n_k\, n_U\, \bar{D}_k(1-\bar{D}_k) \qquad \text{treated vs. never-treated}$$

$$s^k_{kl} \propto n_k\, n_l\, (\bar{D}_k - \bar{D}_l)(1-\bar{D}_k) \qquad \text{early vs. late, before late switches}$$

$$s^l_{kl} \propto n_k\, n_l\, \bar{D}_l(\bar{D}_k - \bar{D}_l) \qquad \text{late vs. early, after early switches}$$

- Every weight has the $\bar{D}(1-\bar{D})$ shape of a Bernoulli variance, so groups treated near the middle of the panel dominate: the weights come from OLS mechanics, not from group sizes alone and not from any estimand you chose. Compare Week 1's variance-weighting result: same phenomenon, new costume.

# The Goodman-Bacon Weights, Explicitly ★ (cont.)

- The forbidden weight $s_{kl}^{l}$ is not negligible in general: it grows with $\bar{D}_k$, i.e. with how long the early group has already been treated when the late group switches.
- All of this is computable from the data alone: `bacondecomp::bacon()` lists every $2 \times 2$, its estimate, and its weight. We decompose a real TWFE estimate in lab.

# Where the Negative Weights Come From ★

FWL (Week 1) reduces TWFE to a bivariate regression on the *double-demeaned* treatment. With $\bar{D}_i$, $\bar{D}_t$, $\bar{D}$ the unit, period, and grand means of $D_{it}$:

$$\hat{\beta}^{\text{TWFE}} = \frac{\sum_{it} \tilde{D}_{it}\, Y_{it}}{\sum_{it} \tilde{D}_{it}^2}, \qquad \tilde{D}_{it} = D_{it} - \bar{D}_i - \bar{D}_t + \bar{D}.$$

Substitute $Y_{it} = \alpha_i + \lambda_t + \tau_{it} D_{it} + \epsilon_{it}$: the fixed effects are annihilated because $\tilde{D}$ has zero unit and period sums, and $\sum_{it} \tilde{D}_{it}^2 = \sum_{(i,t):D_{it}=1} \tilde{D}_{it}$, so

$$\text{plim}\ \hat{\beta}^{\text{TWFE}} = \sum_{(i,t):\, D_{it}=1} w_{it}\, \tau_{it}, \qquad w_{it} = \frac{\tilde{D}_{it}}{\sum_{(i,t):\, D_{it}=1} \tilde{D}_{it}}, \qquad \sum w_{it} = 1.$$

- For a *treated* cell, $\tilde{D}_{it} = 1 - \bar{D}_i - \bar{D}_t + \bar{D}$, which is negative exactly when $\bar{D}_i + \bar{D}_t > 1 + \bar{D}$: a unit treated for most of the panel, observed in a period when most of the sample is treated. That is the early adopter, late in the panel.

# Where the Negative Weights Come From ★ (cont.)

- Intuition: OLS rewards cells whose treatment status is *surprising* given their unit and period averages. An early adopter's late-period treatment is completely unsurprising, so its effect is priced at zero, or below zero.
- The weights are a mechanical fact about OLS and are there even under constant effects, harmlessly. Heterogeneous $\tau_{it}$ is what turns strange weights into a wrong answer.

# Negative Weights: Consequences and Diagnostics ★

de Chaisemartin and D'Haultfœuille (2020) generalize the previous frame's calculation, allowing covariates and non-binary treatments, and show that negative weights are the rule in staggered designs, not a corner case.

- Consequence: TWFE can have the wrong sign even when every $\tau_{it} > 0$. This is not a small-sample problem; it survives $n \to \infty$, because the weights, not the noise, are at fault.
- By the formula, the negative weights fall on units treated *early* and observed over a *long* post-period, exactly the units carrying the most information about long-run effects. TWFE quietly discounts, and then inverts, the long run.

# Negative Weights: Consequences and Diagnostics ★ (cont.)

- **Diagnostic**: compute the weights (`TwoWayFEWeights::twowayfeweights`) and report how many are negative and what share of the total mass they carry. Small negative mass is reassuring about *this particular* problem; it says nothing about parallel trends or about dynamic-effect contamination, so it is one diagnostic, not a validity test.
- Sun and Abraham (2021) show the same contamination in event-study coefficients: with staggered timing, the TWFE $\hat{\tau}_k$ at lag $k$ is a weighted combination of *every* cohort's effects at *every* lag, with weights that need not be positive or sum to one lag by lag. So apparent pre-trends can be artefacts of post-treatment dynamics, and a clean pre-trend plot from raw TWFE proves nothing.

TWFE is fine when treatment timing is common, or when effects are genuinely constant. It is unreliable exactly where sociologists use it most.

# Negative Weights on the Divorce Data

The formula two frames back, computed cell by cell: $w_{it} = \tilde{D}_{it} / \sum_{D_{it}=1} \tilde{D}_{it}$ with $\tilde{D}$ the double-demeaned treatment (figs/w10_divorce.R).

| | |
|---|---|
| treated state-years | 1,164 |
| with a negative weight | 267 |
| sum of the negative weights | −0.38 |
| states carrying them | adopted before 1964, 1969, 1970 |
| years | 1974–1996 |

- Exactly where the formula said: early adopters, late in the panel. Their long-run effects, which Stevenson and Wolfers find are the largest, enter TWFE with the wrong sign.
- Positive weights sum to 1.38, so TWFE is not a convex average of any group's effect.

# Callaway–Sant'Anna: Build Clean Blocks, Then Aggregate

Define the group-time average treatment effect for cohort $g$ at time $t$:

$$ATT(g,t) = E\big[Y_t(g) - Y_t(\infty) \mid G_i = g\big],$$

identified by comparing cohort $g$ to units *not yet treated* at $t$ (or to never-treated units), using only periods before $g$ as the baseline. Under cohort-level parallel trends and no anticipation,

$$ATT(g,t) = E\big[Y_t - Y_{g-1} \mid G_i = g\big] - E\big[Y_t - Y_{g-1} \mid \text{not yet treated at } t\big] :$$

a clean $2 \times 2$. The baseline $g-1$ predates cohort $g$'s treatment, the comparison group has not started treatment by $t$, and no already-treated unit is ever used as a control.

- Under *conditional* parallel trends, each block can be estimated by IPW, outcome regression, or a doubly robust estimator, Week 6, reappearing inside a DID.

# Callaway–Sant'Anna: Build Clean Blocks, Then Aggregate (cont.)

- **Then choose an aggregation**, and the choice is an estimand choice:
  - *simple*: average over all treated $(g, t)$;
  - *dynamic*: average by $t - g$, giving an event-study path;
  - *group*: average within cohort, showing who benefited;
  - *calendar*: average within period, showing when.
- Sun–Abraham reaches the same place through an interaction-weighted regression; Borusyak–Jaravel–Spiess by imputing $Y(0)$ from untreated cells.

# Imputation, Step by Step (Borusyak, Jaravel, and Spiess 2024)

Use only untreated cells to learn the untreated model, then fill in:

1. $Y_{it} = \alpha_i + \lambda_t + \epsilon_{it}$ on cells with $D_{it} = 0$ — never-treated, and pre-periods of the treated
2. $\hat{Y}_{it}(0) = \hat{\alpha}_i + \hat{\lambda}_t$ for cells with $D_{it} = 1$ — impute the missing potential outcome
3. $\hat{\tau}_{it} = Y_{it} - \hat{Y}_{it}(0), \quad \hat{\tau} = \sum_{D_{it}=1} v_{it}\,\hat{\tau}_{it}$ — average with weights you choose

- No treated cell is ever a control; any aggregation ($v_{it}$) is allowed, so the estimand is chosen, not inherited from OLS.
- **Efficient** when errors are homoskedastic and serially uncorrelated: it uses all pre-periods, not just $g - 1$.
- **Pre-trend tests are separate**: run on untreated cells only, so testing does not change the estimate.
- Gardner (2022)'s two-stage DiD is the same idea as two regressions (`did2s`).

# Extended TWFE: the Regression Was Not the Problem

**(Wooldridge 2025)**

TWFE failed because it forced one $\beta$ on all treated cells. Give every cohort–period its own coefficient:

$$Y_{it} = \alpha_i + \lambda_t + \sum_g \sum_{t \geq g} \tau_{gt} \, \mathbb{1}\{G_i = g\} \, \mathbb{1}\{\text{period} = t\} + \epsilon_{it}.$$

- The unit and period effects are then fitted on untreated cells only (every treated cell has its own dummy), so $\hat{\tau}_{gt}$ equals the imputation estimate. Divorce data: both give $-4.85$.
- Adding the pre-period interactions too, against never-treated states, reproduces Sun–Abraham and Callaway–Sant'Anna with a universal base: all three give $-10.56$.
- Covariates enter as cohort-by-period interactions (the "two-way Mundlak" regression). Nonlinear versions (logit, Poisson) follow (`etwfe`).

# Stacked DiD and Its Weights

**Stacking** (Cengiz, Dube, Lindner, and Zipperer 2019): for each adoption cohort, build a sub-experiment (the cohort plus clean controls, in a fixed window around adoption), stack them, and run one regression with sub-experiment-specific unit and time effects.

- Never uses already-treated controls, and is easy to explain.
- Unweighted stacking does not estimate a clean average: each sub-experiment's treated and control shares differ, so OLS weights their trends unequally (Wing, Freedman, and Hollingsworth 2024). Corrective sample weights recover a "trimmed aggregate ATT".
- Used in sociology: Walmart Supercenters and unions (Choper, Lehner, and Parolin 2026); welfare reform and children's home environments (Kalil et al. 2023), both later.

# The Modern Estimators ★

All of them share one principle: never use an already-treated unit as a control. Build clean $2 \times 2$ comparisons, then aggregate with weights you choose.

| Method | Idea | R |
|---|---|---|
| Callaway–Sant'Anna (2021) | group-time ATT$(g, t)$, never- or not-yet-treated controls | `did` |
| Sun–Abraham (2021) | interaction-weighted event study | `fixest::su` |
| de Chaisemartin–D'Haultfœuille | clean comparisons at switch dates | `DIDmultipl` |
| Borusyak–Jaravel–Spiess | impute $Y(0)$ from untreated cells | `didimputat` |
| Stacked / cohort DID | one clean event window per cohort | manual |

Callaway–Sant'Anna also allows **conditional** parallel trends: trends parallel only after adjusting for covariates, estimated by IPW, regression, or doubly robust, Week 6, reappearing.

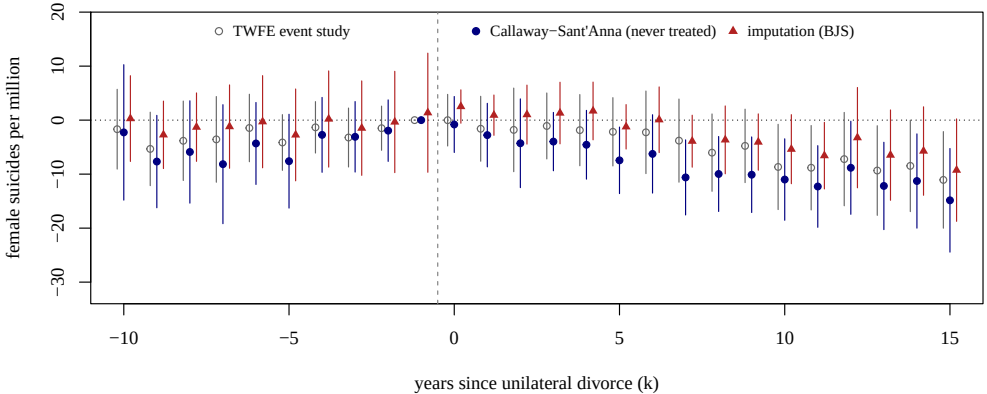
# Six Estimators, One Dataset

Divorce data without the 9 always-treated states (they have no pre-period); standard errors clustered by state (figs/w10_divorce.R):

| estimator | controls, baseline | ATT | (se) |
|---|---|---|---|
| TWFE, all 51 states | everything | −3.05 | (2.32) |
| TWFE, 42 states | everything | −0.34 | (2.76) |
| Callaway–Sant'Anna | never treated, year $g-1$ | −10.56 | (3.12) |
| Callaway–Sant'Anna | not yet treated, year $g-1$ | −8.69 | (3.40) |
| Sun–Abraham | never treated, year $g-1$ | −10.56 | (3.52) |
| imputation = extended TWFE | not yet treated, all pre-years | −4.85 | (2.97) |

- The robust estimators agree on the sign and say TWFE understates the effect.
- They differ in size because they assume parallel trends over different windows: from $g-1$ only, or over every pre-period. Next frame shows why that matters here.

# The Event Studies, Side by Side




- Leads are not flat (Callaway–Sant'Anna at $k = -5$: $-7.6$): untreated trends were not parallel in every pre-year, so "which pre-years define the baseline" changes the answer.
- Effects grow with time since adoption ($-14.9$ at $k = 15$), the dynamic heterogeneity that breaks TWFE.

# Choices Every Staggered DiD Makes

| choice | options | consequence |
|---|---|---|
| comparison group | never treated; not yet treated | not-yet uses more data but needs no anticipation and parallel trends with future adopters |
| baseline | $g-1$ only; all pre-periods | all pre-periods is efficient if trends are parallel throughout, biased if not |
| aggregation | simple; by event time; by cohort; by calendar time | each is a different estimand; say which |
| anticipation | shift $g$ back by $\delta$ periods | otherwise effects leak into the baseline |
| covariates | none; DR within each $ATT(g,t)$ | use pre-treatment values only |
| inference | cluster; bootstrap; randomization | few treated cohorts need care |

Efficiency when adoption timing is as good as random: Roth and Sant'Anna (2023) (`staggered`). A design-based reading of TWFE under random timing: Athey and Imbens (2022).

# Honest Inference About Pre-Trends

Rather than testing pre-trends and proceeding as if they had passed, Rambachan and Roth (2023) propose bounding how badly parallel trends could fail.

- Their "relative magnitudes" restriction bounds how much the *change* in the trend violation after treatment can exceed the largest change observed before it, by a factor $\bar{M}$, and the resulting confidence sets account for sampling uncertainty in the pre-period estimates rather than treating them as known.
- Report the *breakdown value*: how large $\bar{M}$ would have to be before your conclusion reverses.
- This turns an untestable assumption into a quantitative claim, exactly the move sensitivity analysis made in Week 6.
- `HonestDiD`.

# Honest Inference About Pre-Trends (cont.)

## Inference reminder

Bertrand, Duflo, and Mullainathan (2004): cluster on the unit at which treatment varies. With few clusters, use CR2 or a wild cluster bootstrap (Week 2).

# Sensitivity on the Divorce Data

Callaway–Sant'Anna event study, universal base; target: the average effect over $k = 0, \ldots, 10$, estimated at $-6.53$ (figs/w10_divorce.R, `HonestDiD`).

| allowed post-period violation, relative to the largest pre-period one ($\bar{M}$) | 95% interval |
|---|---|
| none (parallel trends exact) | $[-11.9, -1.1]$ |
| $\bar{M} = 0.05$ | $[-13.4, 0.8]$ |
| $\bar{M} = 0.25$ | $[-28.4, 16.4]$ |
| $\bar{M} = 1$ | $[-55.1, 55.1]$ |

- Breakdown below $\bar{M} = 0.05$: allowing post-period violations one-twentieth the size of the largest pre-period swing erases the sign.
- The pre-period swings are large (leads near $-7.6$), so relative magnitudes give little room. The estimate is honest about what these data can say.

# Beyond

# Doses: a Continuous Treatment

Callaway, Goodman-Bacon, and Sant'Anna (2025). Units receive dose $d$ (minimum-wage increase in dollars, hours of a program); some get $d = 0$. Two different questions:

$$\mathrm{ATT}(d \mid d) = E[Y_t(d) - Y_t(0) \mid D = d] \qquad \text{effect of dose } d \text{ on those who got } d$$

$$\mathrm{ATT}(d \mid d) - \mathrm{ATT}(d' \mid d') = \underbrace{\mathrm{ATT}(d \mid d) - \mathrm{ATT}(d' \mid d)}_{\text{causal: more dose, same units}} + \underbrace{\mathrm{ATT}(d' \mid d) - \mathrm{ATT}(d' \mid d')}_{\text{selection: different units}}$$

- Ordinary parallel trends (each dose group vs. $d = 0$) identifies each $\mathrm{ATT}(d \mid d)$, but not how effects change with the dose.
- Comparing doses needs **strong parallel trends**: every dose group would have trended like any other under every dose. It rules out selection on gains.
- TWFE on a continuous $D$ mixes both terms with weights that do not follow the dose distribution; its weights on level effects sum to zero (`contdid`).

# Treatments That Switch On and Off

Union membership, unemployment, living with a partner: units enter and leave treatment, and past treatment may still matter.

- TWFE with $D_{it}$ switching assumes no carryover (Imai–Kim) and constant effects, and uses switchers as each other's controls.
- de Chaisemartin and D'Haultfœuille (2026): compare units whose treatment first changes at $t$ with units that had the *same* treatment history up to $t$ and have not yet changed. The effect of having been exposed for $\ell$ periods, $\text{DID}_\ell$, is an event study for non-absorbing, non-binary treatments (`DIDmultiplegtDYN`).
- Matching on treatment histories (Imai, Kim, and Wang 2023) does the same with explicit lag length $L$.

Cost: only units with a matching history count, so fewer comparisons and wider intervals. That is the price of not assuming carryover away.

# Triple Differences

A policy hits one group (say, mothers) in treated states. Difference the state DiD for mothers against the state DiD for a group the policy should not touch (women without children):

$$\text{DDD} = \text{DiD}_{\text{mothers}} - \text{DiD}_{\text{non-mothers}}.$$

- Needs only **one** parallel-trends assumption: the bias in the two DiDs is the same. Neither DiD needs to be unbiased on its own (Olden and Møen 2022).
- With covariates or staggered timing, "the difference of two DiDs" and three-way fixed effects regressions are generally not valid; pooling not-yet-treated units can add bias even without covariates. Use estimators built for DDD (Ortiz-Villavicencio and Sant'Anna 2025; `tripledifff`).

# Distributions, and a Design-Based View

**Effects on the distribution** (Callaway and Li 2019): with three or more panel periods, distributional parallel trends plus copula stability (the link between a unit's change in $Y(0)$ and its level stays the same over time) identify quantile effects on the treated.

**Where does uncertainty come from?** Standard errors here usually imagine resampling units. In policy panels we observe every state.

- Athey and Imbens (2022): treat the *adoption date* as the random element; TWFE then has a design-based reading, and the usual variance is conservative.
- Rambachan and Roth (2026): design-based uncertainty for quasi-experiments; usual standard errors can remain valid, if conservative, for a well-defined estimand.

**Factor models** (unit-specific responses to common shocks) relax parallel trends further: synthetic DiD, generalized synthetic control, counterfactual estimators (`fect`). Next week.

# Sociology

# In the Literature: Harknett, Schneider, and Irwin (2021)

**The question.** Does making work schedules predictable improve workers' lives? Seattle's Secure Scheduling Ordinance (2017): two weeks' notice, pay for last-minute changes, extra pay for "clopening" shifts; large retail and food-service firms.

**The data.** The Shift Project: surveys of hourly workers recruited by Facebook and Instagram ads. Seattle: 754 workers at baseline (spring 2017), 747 after one year, 441 after two. Comparison: workers at the *same firms* in 24 large metros with above-federal minimum wages. 17,689 responses in all.

**The design.** DiD with two post periods,

$$Y_{ict} = \beta_1 \,\text{Seattle}_c \times \text{Year1}_t + \beta_2 \,\text{Seattle}_c \times \text{Year2}_t + \text{city and wave effects} + X'_{ict}\delta + \epsilon_{ict},$$

linear probability models, weighted to the American Community Survey. Same firms on both sides holds employer practices fixed, as Card and Krueger held the industry fixed.

# Harknett, Schneider, and Irwin: Checking the Design

| | |
|---|---|
| Baseline levels | Similar: unpredictability scale 2.79 vs. 2.85; happy 76% vs. 75%; any material hardship 60% vs. 58%. Clopening differs (37% vs. 44%). |
| Pre-trends | No pre-2017 data, so no event study. Instead: in places without scheduling laws, the national scale stays near 2.8 in 2017, 2018, 2019. |
| Comparison group | Results hold with cities that considered but did not pass a law, with a propensity-matched sample, and (smaller) with workers just outside Seattle, where spillover is plausible. |
| Composition | The Seattle sample becomes slightly younger and shorter-tenured. |
| Sampling | Not a probability sample; more women and White workers than the workforce. |

A one-treated-city DiD: inference rests on the comparison cities behaving like Seattle would have. Next week's synthetic control is built for exactly this case.

# Harknett, Schneider, and Irwin: Main Results

Change in Seattle minus change in comparison cities after the law (their Figures 1–2):

| outcome | DiD estimate |
|---|---|
| schedule unpredictability scale (baseline 2.8) | $-0.50$ $(-18\%)$ |
| at least two weeks' notice | $+11$ pts |
| last-minute changes without pay | $-13$ pts |
| on-call shifts | $-7$ pts $(p \leq 0.07)$ |
| happy | $+7$ pts |
| good sleep quality | $+11$ pts |
| any material hardship | $-10$ pts |
| psychological distress | n.s. |

The law changed schedules first; well-being followed. An IV extension uses the law as an instrument for predictability (first-stage $F = 17.1$).

# Staggered DiD in Sociology: Two Recent Studies

**Choper, Lehner, and Parolin (2026, ASR)**: Walmart Supercenters and unions. Restricted PSID 1983–2019; 966 first Supercenter openings across counties; stacked DiD with never-treated controls, matched workers, ±10 years.

- Union membership falls 1.4 points (se 0.8); 3.5 points (1.3) when differential linear pre-trends are removed: about 21% of the 16.1% baseline.
- The headline depends on a trend adjustment: report both, as they do.

**Kalil, Corman, Dave, Schwartz-Soicher, and Reichman (2023, *Demography*)**: staggered state welfare reform 1992–98; NLSY79 unmarried, low-educated mothers of young children.

| home environment score | TWFE | stacked (not-yet-treated controls) |
|---|---|---|
| emotional support | −4.95 (1.44) | −6.56 (1.77) |
| overall | −2.89 | −5.40 |

Already-treated controls attenuate TWFE when effects grow, as in the divorce data.

# Practice

# What a Reanalysis of 49 Studies Found (Chiu, Lan, Liu, and Xu 2026)

Political scientists reran 49 published panel studies with the heterogeneity-robust estimators of this lecture.

- The new estimators usually agree with TWFE *qualitatively*, but the estimates vary a lot across estimators.
- Many studies lack evidence for parallel trends once pre-periods are examined properly.
- Many are underpowered: the robust estimators' intervals are wide.

Read with the divorce data in mind: the estimator mattered (TWFE vs. robust), and so did the window over which trends are assumed parallel. The practical guide that collects these steps: Baker, Callaway, Cunningham, Goodman-Bacon, and Sant'Anna (2026).

# A DID Checklist ★

1. State the estimand: ATT for whom, over which post-treatment window?
2. Plot the raw group means over time. Before any regression.
3. Is timing staggered? If so, do not use plain TWFE. Use a modern estimator, and report the TWFE estimate only for comparison.
4. Report an event-study plot with pre-period coefficients, using Sun–Abraham or Callaway–Sant'Anna rather than raw TWFE leads.
5. Justify the scale (levels or logs) on substantive grounds.
6. Cluster on the treatment-assignment unit; report the number of clusters.
7. Report a sensitivity analysis for parallel trends.
8. Placebo: a pre-treatment outcome, or an untreated outcome that the policy should not affect.

# Looking Ahead

- Fixed effects difference away time-invariant confounders, a real gain, bought with precision and with a new assumption about trends.
- The staggered-adoption literature is a case study in how a method can be standard for decades and still be wrong. Read papers published before 2021 with this in mind, including your own field's classics.
- **Next week**: what to do when there is only *one* treated unit, and "the control group" has to be constructed.
- Notice that conditional parallel trends brought Week 6 back inside a DID: once trends are only parallel given covariates, you need a weighting or outcome model again, and all of Week 6's cautions return with it.

**Problem Set 4**

Assigned today, due Monday November 9. Covers Weeks 7 and 9.

# References


- Abadie, Alberto. 2005. "Semiparametric Difference-in-Differences Estimators." *Review of Economic Studies* 72(1): 1–19.
- Allison, Paul D. 2009. *Fixed Effects Regression Models*. Sage.
- Anderson, T. W. and Cheng Hsiao. 1981. "Estimation of Dynamic Models with Error Components." *Journal of the American Statistical Association* 76(375): 598–606.
- Angrist, Joshua D. and Jörn-Steffen Pischke. 2009. *Mostly Harmless Econometrics*. Princeton University Press. [Ch. 5]
- Arellano, Manuel. 1987. "Computing Robust Standard Errors for Within-Groups Estimators." *Oxford Bulletin of Economics and Statistics* 49(4): 431–434.
- Arellano, Manuel and Stephen Bond. 1991. "Some Tests of Specification for Panel Data: Monte Carlo Evidence and an Application to Employment Equations." *Review of Economic Studies* 58(2): 277–297.
- Athey, Susan and Guido W. Imbens. 2022. "Design-Based Analysis in Difference-in-Differences Settings with Staggered Adoption." *Journal of Econometrics* 226(1): 62–79.

# References (cont.)


- Baker, Andrew, Brantly Callaway, Scott Cunningham, Andrew Goodman-Bacon, and Pedro H.C. Sant'Anna. 2026. "Difference-in-Differences Designs: A Practitioner's Guide." *Journal of Economic Literature* 64(2): 498–557.
- Bertrand, Marianne, Esther Duflo, and Sendhil Mullainathan. 2004. "How Much Should We Trust Differences-in-Differences Estimates?" *Quarterly Journal of Economics* 119(1): 249–275.
- Borusyak, Kirill, Xavier Jaravel, and Jann Spiess. 2024. "Revisiting Event-Study Designs: Robust and Efficient Estimation." *Review of Economic Studies* 91(6): 3253–3285.
- Brüderl, Josef and Volker Ludwig. 2015. "Fixed-Effects Panel Regression." Pp. 327–357 in *The SAGE Handbook of Regression Analysis and Causal Inference*, edited by H. Best and C. Wolf. Sage.
- Budig, Michelle J. and Paula England. 2001. "The Wage Penalty for Motherhood." *American Sociological Review* 66(2): 204–225.
- Caetano, Carolina and Brantly Callaway. 2024. "Difference-in-Differences when Parallel Trends Holds Conditional on Covariates." arXiv:2406.15288.

# References (cont.)


- Callaway, Brantly, Andrew Goodman-Bacon, and Pedro H.C. Sant'Anna. 2025. "Difference-in-Differences with a Continuous Treatment." arXiv:2107.02637 (NBER WP 32117).
- Callaway, Brantly and Tong Li. 2019. "Quantile Treatment Effects in Difference in Differences Models with Panel Data." *Quantitative Economics* 10(4): 1579–1618.
- Callaway, Brantly and Pedro H.C. Sant'Anna. 2021. "Difference-in-Differences with Multiple Time Periods." *Journal of Econometrics* 225(2): 200–230.
- Card, David and Alan B. Krueger. 1994. "Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania." *American Economic Review* 84(4): 772–793.
- Cengiz, Doruk, Arindrajit Dube, Attila Lindner, and Ben Zipperer. 2019. "The Effect of Minimum Wages on Low-Wage Jobs." *Quarterly Journal of Economics* 134(3): 1405–1454.
- Chiu, Albert, Xingchen Lan, Ziyi Liu, and Yiqing Xu. 2026. "Causal Panel Analysis under Parallel Trends: Lessons from a Large Reanalysis Study." *American Political Science Review* 120(1): 245–266.

# References (cont.)


- Choper, Joshua, Clemens Lehner, and Zachary Parolin. 2026. "Countervailing Powers: Labor Unions Against the Buyer Power of Walmart Supercenters." *American Sociological Review* 91(3): 435–463.
- Conley, Timothy G. and Christopher R. Taber. 2011. "Inference with 'Difference in Differences' with a Small Number of Policy Changes." *Review of Economics and Statistics* 93(1): 113–125.
- de Chaisemartin, Clément and Xavier D'Haultfœuille. 2020. "Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects." *American Economic Review* 110(9): 2964–2996.
- de Chaisemartin, Clément and Xavier D'Haultfœuille. 2026. "Difference-in-Differences Estimators of Intertemporal Treatment Effects." *Review of Economics and Statistics* 108(4): 863–880.
- Freyaldenhoven, Simon, Christian Hansen, Jorge Pérez Pérez, and Jesse M. Shapiro. 2025. "Visualization, Identification, and Estimation in the Linear Panel Event-Study Design." Pp. 225–268 in *Advances in Economics and Econometrics: Twelfth World Congress*, Vol. 2. Cambridge University Press.

# References (cont.)


- Freyaldenhoven, Simon, Christian Hansen, and Jesse M. Shapiro. 2019. "Pre-Event Trends in the Panel Event-Study Design." *American Economic Review* 109(9): 3307–3338.
- Gardner, John. 2022. "Two-Stage Differences in Differences." arXiv:2207.05943.
- Goodman-Bacon, Andrew. 2021. "Difference-in-Differences with Variation in Treatment Timing." *Journal of Econometrics* 225(2): 254–277.
- Harknett, Kristen, Daniel Schneider, and Véronique Irwin. 2021. "Improving Health and Economic Security by Reducing Work Schedule Uncertainty." *Proceedings of the National Academy of Sciences* 118(42): e2107828118.
- Heckman, James J., Hidehiko Ichimura, and Petra E. Todd. 1997. "Matching as an Econometric Evaluation Estimator: Evidence from Evaluating a Job Training Programme." *Review of Economic Studies* 64(4): 605–654.
- Hausman, Jerry A. 1978. "Specification Tests in Econometrics." *Econometrica* 46(6): 1251–1271.

# References (cont.)


- Imai, Kosuke and In Song Kim. 2019. "When Should We Use Unit Fixed Effects Regression Models for Causal Inference with Longitudinal Data?" *American Journal of Political Science* 63(2): 467–490.
- Imai, Kosuke, In Song Kim, and Erik H. Wang. 2023. "Matching Methods for Causal Inference with Time-Series Cross-Sectional Data." *American Journal of Political Science* 67(3): 587–605.
- Kalil, Ariel, Hope Corman, Dhaval M. Dave, Ofira Schwartz-Soicher, and Nancy E. Reichman. 2023. "Welfare Reform and the Quality of Young Children's Home Environments." *Demography* 60(6): 1791–1813.
- Kleven, Henrik, Camille Landais, and Jakob Egholt Søgaard. 2019. "Children and Gender Inequality: Evidence from Denmark." *American Economic Journal: Applied Economics* 11(4): 181–209.
- Ludwig, Volker and Josef Brüderl. 2018. "Is There a Male Marital Wage Premium? New Evidence from the United States." *American Sociological Review* 83(4): 744–770.
- Ludwig, Volker and Josef Brüderl. 2021. "What You Need to Know When Estimating Impact Functions with Panel Data for Demographic Research." *Comparative Population Studies* 46.

# References (cont.)


- MacKinnon, James G. and Matthew D. Webb. 2018. "The Wild Bootstrap for Few (Treated) Clusters." *Econometrics Journal* 21(2): 114–135.
- Mundlak, Yair. 1978. "On the Pooling of Time Series and Cross Section Data." *Econometrica* 46(1): 69–85.
- Nickell, Stephen. 1981. "Biases in Dynamic Models with Fixed Effects." *Econometrica* 49(6): 1417–1426.
- Olden, Andreas and Jarle Møen. 2022. "The Triple Difference Estimator." *Econometrics Journal* 25(3): 531–553.
- Ortiz-Villavicencio, Marcelo and Pedro H.C. Sant'Anna. 2025. "Better Understanding Triple Differences Estimators." arXiv:2505.09942.
- Rambachan, Ashesh and Jonathan Roth. 2023. "A More Credible Approach to Parallel Trends." *Review of Economic Studies* 90(5): 2555–2591.
- Rambachan, Ashesh and Jonathan Roth. 2026. "Design-Based Uncertainty for Quasi-Experiments." *Journal of the American Statistical Association* 121(553): 477–491.

# References (cont.)


- Roodman, David. 2009. "A Note on the Theme of Too Many Instruments." *Oxford Bulletin of Economics and Statistics* 71(1): 135–158.
- Roth, Jonathan. 2022. "Pretest with Caution: Event-Study Estimates after Testing for Parallel Trends." *American Economic Review: Insights* 4(3): 305–322.
- Roth, Jonathan. 2026. "Interpreting Event-Studies from Recent Difference-in-Differences Methods." *Japanese Economic Review* 77(2): 275–288.
- Roth, Jonathan and Pedro H.C. Sant'Anna. 2023. "When Is Parallel Trends Sensitive to Functional Form?" *Econometrica* 91(2): 737–747.
- Roth, Jonathan and Pedro H.C. Sant'Anna. 2023. "Efficient Estimation for Staggered Rollout Designs." *Journal of Political Economy Microeconomics* 1(4): 669–709.
- Roth, Jonathan, Pedro H.C. Sant'Anna, Alyssa Bilinski, and John Poe. 2023. "What's Trending in Difference-in-Differences?" *Journal of Econometrics* 235(2): 2218–2244.

# References (cont.)


- Rüttenauer, Tobias and Volker Ludwig. 2023. "Fixed Effects Individual Slopes: Accounting and Testing for Heterogeneous Effects in Panel Data or Other Multilevel Models." *Sociological Methods & Research* 52(1): 43–84.
- Sant'Anna, Pedro H.C. and Jun Zhao. 2020. "Doubly Robust Difference-in-Differences Estimators." *Journal of Econometrics* 219(1): 101–122.
- Stevenson, Betsey and Justin Wolfers. 2006. "Bargaining in the Shadow of the Law: Divorce Laws and Family Distress." *Quarterly Journal of Economics* 121(1): 267–288.
- Sun, Liyang and Sarah Abraham. 2021. "Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects." *Journal of Econometrics* 225(2): 175–199.
- Vella, Francis and Marno Verbeek. 1998. "Whose Wages Do Unions Raise? A Dynamic Model of Unionism and Wage Rate Determination for Young Men." *Journal of Applied Econometrics* 13(2): 163–183.
- Wing, Coady, Seth M. Freedman, and Alex Hollingsworth. 2024. "Stacked Difference-in-Differences." NBER Working Paper 32054.

# References (cont.)


- Wooldridge, Jeffrey M. 2010. *Econometric Analysis of Cross Section and Panel Data*. 2nd ed. MIT Press.
- Wooldridge, Jeffrey M. 2025. "Two-Way Fixed Effects, the Two-Way Mundlak Regression, and Difference-in-Differences Estimators." *Empirical Economics* 69(5): 2545–2587.